Imperial College London

Advanced Signal Processing

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Nice layout of the report, however, there is some inconsistency in section labeling. In general figures are well formatted but some legends are missing or too small. Some statements are not fully explained.

Good title page

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1 Random signals and stochastic processes

The first section of the coursework is focused on characterisation of different signals as being ergodic or stationary based on their statistical properties. Furthermore in order to completely understand the signal's nature, its probability density function was also investigated.

1.1 Statistical estimation

In order to generate a 1000-sample vector the *randn* function was used, generating random and uniformly distributed numbers in the range of 0 to 1. Figure 1 illustrates the plotted realisation of the statistically stationary signal.

section numbering is inconsistent

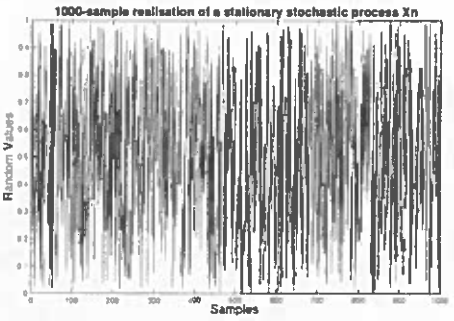


Figure 1: Uniform distribution of 1000 samples

To investigate and calculate the bias in the estimation we generated ten 1000-sample realizations of X and calculated the sample means and sample standard deviations for each one of them. The mean and standard deviation were computed using the $MATLAB$ commands $mean(\hat{m})$ and $std(\hat{\sigma})$ which are based the equation below.

good format of equations

$$\hat{m} = \frac{1}{N}\sum_{n}^{N} x[n] \qquad \hat{\sigma} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}(x[n]-\hat{m})^2} \tag{1}$$

It should be noted that the theoretical mean(E[x]) and standard deviation(std) is based on the following equations:

$$E[X] = \int_a^b x f(x)dx = \int_a^b x\frac{1}{b-a}dx = \frac{b+a}{2} \qquad std = \frac{(b-a)^2}{12} = \frac{1}{12} \tag{2}$$

Having in mind the difference between the two approaches the mean and standard deviation was computed and the deviation from the theoretical values was noted. $MATLAB$ calculated the *mean* and *std* to be 0.4988 and 0.2915 respectively. The theoretical values were calculated to be 0.5 for the *mean* and 0.2887 for the *std*. Even though the sample mean is an efficient and unbiased estimator it didn't appear to produce the same value as the theoretical mean. This occurred because when different sample ranges are chosen from the same distribution, different sample means would be produced and therefore different estimates. The estimator itself exhibits a variability since it is a random variable with mean μ and variance σ^2/N.The percentage error for the mean and std estimators were found to be 2.81% and 0.9865%

respectively.
3.After generating ten 1000-sample realisations of the uniform random variable $X \sim U(0,1)$ the sample means and stds were plotted.

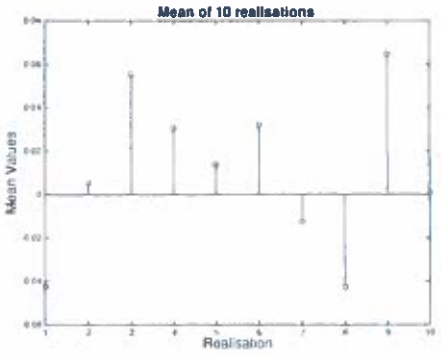


Figure 2: Sample mean of 10 realisations

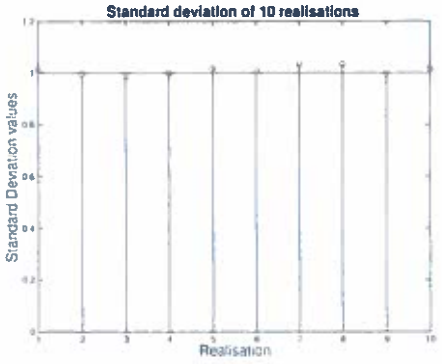


Figure 3: Sample std of 10 realisations

	Mean ($\hat{m}$)	Standard deviation ($\hat{\sigma}$)
Theoretical statistic	0.5	0.2887
$\mathbb{E}(\hat{m})$ or $\mathbb{E}(\hat{\sigma})$	0.5049	0.2877
$\mathbb{E}(bias)$	-0.0051	0.0021
Range	0.0357	0.0092

Good use of the table and correct results

Table 1: Statistical properties for 10 $U|0,1|$ realisations

As it is highlighted from figure (2) and (3), the aforementioned predicted variability is confirmed. A fluctuation in the values of the mean and std for the ten realisations is evident.
4.For a more complete conclusion regarding their stochastic description, the probability density function was investigated. in part 1.1.4 the pdf of the randomly generated signal was approximated using $MATLAB$. The results are depicted below:

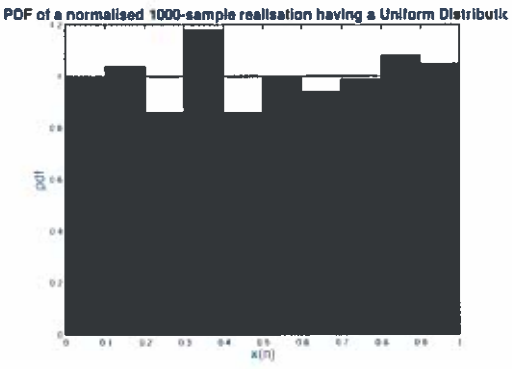


Figure 4: PDF of 1000-sample realisation of X

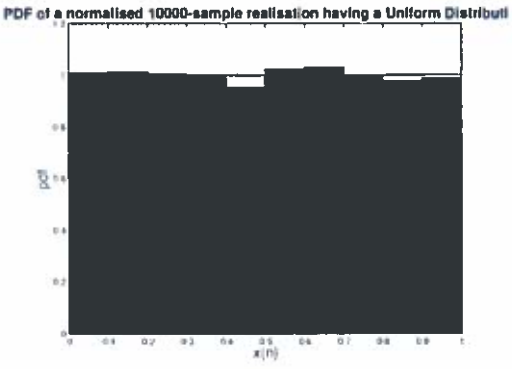


Figure 5: PDF of 10000-sample realisation of X

Good captions

The red line in figure(4) shows the theoretical pdf of a uniform distribution. The blue bars represent the the approximations. As it is observed, the approximated PDF becomes increasingly more accurate as the number of samples generated is increased from 100 to 10000. This can be seen on figure (5), where as the number of samples increases the converge more towards the red line representing the theoretical value. 5.The process was repeated to generate a zero-mean, unit standard deviasion, Gaussian distribution, X N(0,1). In order to obtain the results, the *MATLAB* function *randn* was used. The sample mean was also calculated using *MATLAB* to be $m = 0.030$. The theoretical mean for a normal Gaussian distribution and theoretical standard deviation is known to be $m = 0$ and $\sigma = 1$ respectively. Using MATLAB the mean was computed to be $m = 0.0082$ and the standard deviation $\sigma = 1.0272$. Figure (6) and figure (7) depict the mean and standard deviation of an ensemble of ten 1000-sampled realisations of X.

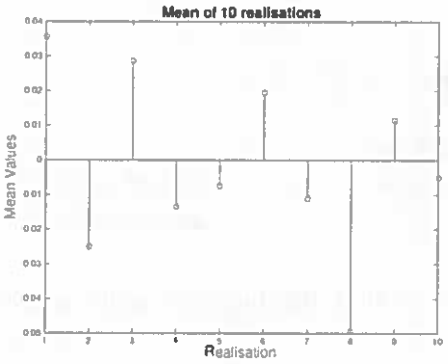


Figure 6: Sample mean of 10 realisations of X

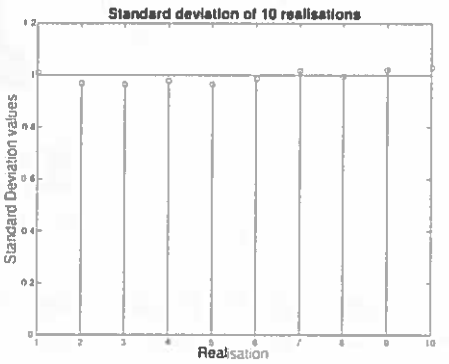


Figure 7: Sample std of 10 realisations of X

The sample and theoretical values don't have the same value because a finite number of samples is used in the calculation. For the calculation of the approximate *pdf* a *MATLAB* for-loop was created to investigate how different bin numbers and different sample numbers affect the approximation of the *pdf*.

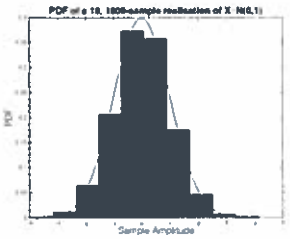


Figure 8: PDF of 1000-sample of X with bin size 5

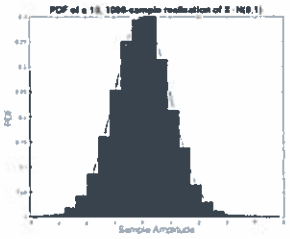


Figure 9: PDF of 1000-sample of X with bin size 10

waste of space

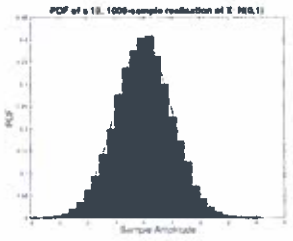


Figure 10: PDF of 1000-sample of X with bin size 20

Not enough of explanation

Looking at figure(9) and figure(10) it is clear that the higher sample number gives a better approximate model for the pdf.

1.2 Stochastic processes

1.Using M=100 and N=100, the ensemble means and standard deviations for each process was calculated and plotted as a function of time.

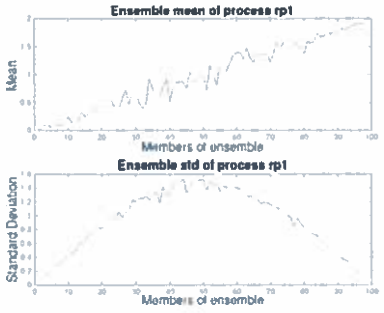


Figure 11: Ensemble statistics of rp1

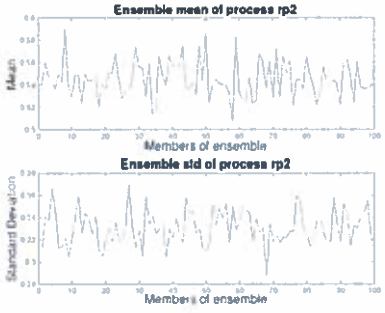
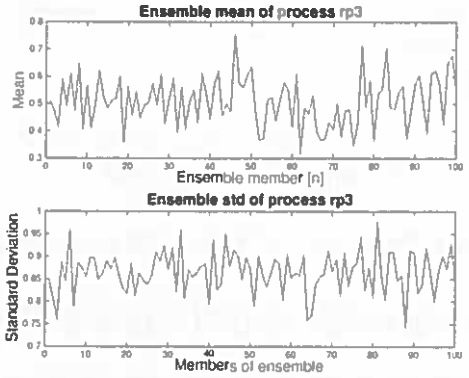


Figure 12: Ensemble statistics of rp2

Correct plots but the available space could've been used better. Also axis could've been scaled better



No definition of stationarity is given

Figure 13: Ensemble statistics of rp3

The first process illustrated at figure(11) is not a stationary process. Both the mean and the standard deviation are varying with time, therefore the stochastic process is a time varying process. The second process illustrated at figure(12) appears to be stationary. The mean and standard deviation is slightly varying but mostly appears to be constant with time incrimination. This variation occured due to the error resulting from the relatively small sample number. Comparably, the third process illustrated at figure(13) can be considered stationary. However, the error in this case is larger resulting to a greater variation of the mean and standard deviation with respect to time(s).

2.In this section of the exercise the ergodicity of each process was investigated using M=4 and N=1000. Ergodicity implies a process in which every sequence or sizable sample is equally representative of the whole and the statistical properties of the process are not changing with time.

Random Process 1				
Mean	10.0675	9.9762	10.0018	10.0211
Standard Deviation	5.9170	5.8777	5.8445	5.8533
Random Process 2				
Mean	0.0375	0.4891	0.7450	0.2062
Standard Deviaton	0.2752	0.0112	0.0622	0.1914
Random Process 3				
Mean	0.4797	0.4491	0.4995	0.5383
Standard Deviation	0.8792	0.8679	0.8778	0.8771

No definition of ergodicity.

Incorrect identification of ergodicity

Table 2: Mean and standard deviation for stochastic processes rp1,rp2 and rp3

Stochastic process 1 doesn't satisfy the criteria to be ergodic because it was found to be a non-stationary process. The values for the mean and standard deviation of the 4 realisations of the Stochastic process three was found to be relatively the same, therefore it may be regarded as ergodic. On the other hand, in stochastic process two, the values for the ensemble and the time statistics are varying. Thus it was concluded that the process is not ergodic. Process two was further investigated with bigger sample numbers and it still displayed non-ergodicity.

3. To write a mathematical description for each of the three stochastic processes, the properties of linearity, separability and uncorrelated random variables were used.

Furthermore, X is considered to be a uniformly distributed random variable with mean=0.5 and variance=1/12.

Process 1: Given: N=1000

$$X_1 \sim U[0,1] \Rightarrow m = 0.5 \text{ and } Var(m) = 1/12.$$

$$Y_1(n) = (X_1 - 0.5)b\sin(\tfrac{n\pi}{N}) + an \qquad X_1 \sim U[0,1], a = 0.02, b = 5$$

$$E\{Y_1(n)\} = E\{(X_1(n)-0.5)b\sin(\frac{n\pi}{N})+an\}E\{(X_1(n)-0.5)b\sin(\frac{n\pi}{N})\}+E\{an\} = E\{X_1(n)b\sin(\frac{n\pi}{N})\} \tag{3}$$

Once the calculation is performed the theoretical mean=10.01.Regarding the std the following calculation was made rendering $Var[y] = 1.44$

$$Var\{Y_1(n)\} = Var\{(X_1(n) - 0.5)b\sin(\frac{n\pi}{N}) + an\}=(b\sin(\frac{n\pi}{N}))^2 Var\{(X_1(n) - 0.5)\}=$$

$$(b\sin(\frac{n\pi}{N}))^2 Var\{X_1(n)\} \overset{given Var(X_1)=1/12}{=} \frac{1}{12}(b\sin(\frac{n\pi}{N}))^2$$

$$\sigma = \sqrt{Var\{Y_1\}} = \sqrt{\frac{1}{12}(b\sin(\frac{n\pi}{N}))^2} = 1.44$$

Process 2: Given: N=1000

m=0.5 Var=1/12 $Y_2(n) = (X_r - 0.5)M_r + A_r \qquad X_r, M_r, A_r \sim U[0,1]$

$$E\{Y_2(n)\} = E\{(X_r(n) - 0.5)M_r + A_r\} = E\{(X_r(n) - 0.5)M_r\} + E\{A_r\} =$$

$$E\{X_r M_r\} - 0.5E\{M_r\} + E\{A_r\} \overset{=}{E}\{X_r\}E\{A_r\} - 0.5E\{M_r\} + E\{A_r\} \overset{given E\{X_r\}=0.5}{=} E\{A_r\} = 0.5$$

$$Var\{Y_2(n)\} = E\{Y_2^2(n)\} - E\{Y_2(n)\}^2 = E\{(X_r M_r - 0.5M_r + A_r)^2\} - 0.5^2 =$$

$$E\{(X_r^2 M_r^2 - X_r M_r^2 + 2X_r M_r A_r - M_r A_r + 0.25M_r^2 + A_r^2\} - 0.25 \overset{given E\{(X_r\}=E\{(M_r\}=E\{(A_r\}=1/3}{=}$$

$$\frac{1}{3}\frac{1}{3} - \frac{1}{2}\frac{1}{3} + 2\frac{1}{2}\frac{1}{2}\frac{1}{2} - \frac{1}{2}\frac{1}{2} + \frac{1}{4}\frac{1}{3} + \frac{1}{3} - \frac{1}{4} = \frac{1}{9} - \frac{1}{6} + \frac{1}{12} + \frac{1}{3} - \frac{1}{4} = \frac{1}{9}$$

$$\sigma = \sqrt{Var\{Y_2\}} = \frac{1}{3}$$

Process 3: Given: N=1000

X_r is a uniformly distributed random variable $\Rightarrow m = 0.5$ and $Var = 1/12$.

$Y_3(n) = (X_r - 0.5)m + a \qquad X_r, X_r \sim U[0,1], m = 3, , a = 0.5$

$$E\{Y_3(n)\} = E\{(X_r(n) - 0.5)m + a\} \overset{=}{m} E\{(X_r(n)\} - 0.5m + a \overset{given E\{X_r\}=0.5}{=} a = 0.5$$

$$Var\{Y_3(n)\} = Var\{(X_r(n) - 0.5)3{+}0.5\} = Var\{(3X_r(n)1\} \overset{=}{9} Var\{X_r\} \overset{Var\{X_r\}=1/12}{=} \frac{9}{12}$$

$$\sigma = \sqrt{Var\{Y_3\}} = \frac{3}{\sqrt{12}} = 0.866$$

equaiton is out of general text margins

Obtained values are correct, but there is no comparison with sample mean and std

1.3 Dealing with computational complexity: sign algorithms

The task of this part is to create and use a pdf since it contains all the statistical information about a random variable.
1. A *MATLAB* function *pdf* was created which employed the *hist* build in function and normalised the sample frequency per area, results of the first task are illustrated in figure(14) and (15)

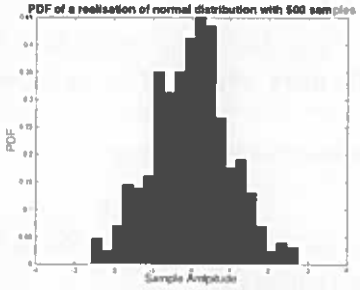


Figure 14: PDF of 100,1000-sample realisation of Normal distribution

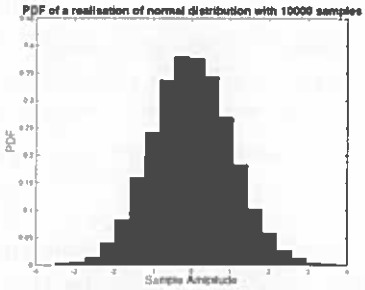


Figure 15: PDF of 100,10000-sample realisation of Normal distribution

No code is given. theoretical pdf is not presented

2. Looking at the previous result only signal rp3 proved to be stationary and ergodic. The pdf was plotted for different data lengths of sample length [100, 1000, 10000]. It was expected that an increase in the number of samples would enhance the convergence of x(n)

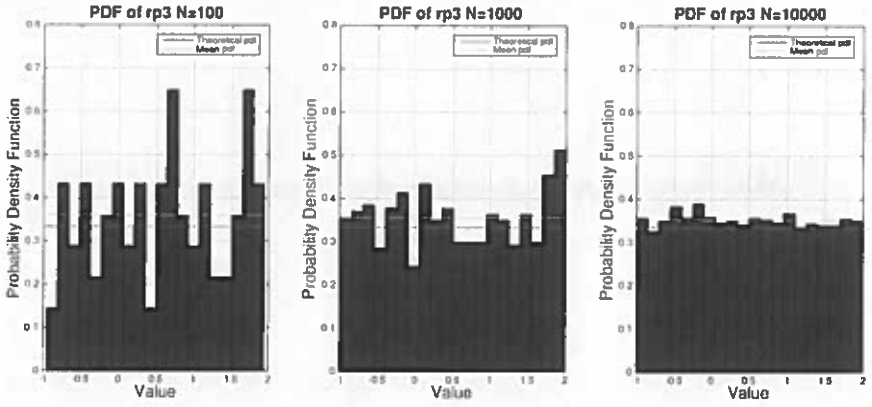


Figure 16: PDF of rp3 N=100,N=1000 and N=10000

Good and correct plots but the legend is small

this part is basically missing

Upon looking at figure (16) it was concluded that by increasing the number of samples, the pdf becomes increasingly more accurate upon comparison with the theoretical one which is equal to $(1/3)$. The results are expected given that an ergodic process can be fully characterised if a signal is sufficiently long.
3. By definition the *pdf* function is only able to estimate the probability density of a stationary process because its statistical properties are time-invariant. As a result, when random process one was observed, it was seen that it wasn't possible to approximate it using the *pdf* function since it didn't display stationarity. When N

was set to 500 the mean appeared different from the time mean of N=1000, leading to the conclusion that the results are not usable.

2 Linear stochastic modelling

2.1 ACF of uncorrelated sequences

1. The fact that the autocorrelation function is a Dirac function indicates that the samples its consists of are uncorrelated thus independent. As a result no information can be extracted looking at the previous or the next sample. This is expressed as:

$$R_x[n, n+\tau] = \begin{cases} E\{X, Xn+\tau\}, & \tau \neq 0 \\ E\{X_n^2,\}, & \tau = 0 \end{cases}$$

In this exercise we are examining the case of an ACF of a WGN. What we expect based on the theory of random independent variables is :

$$R_x[n, n+\tau] = \begin{cases} 0, & \tau \neq 0 \\ \sigma^2, & \tau = 0 \end{cases} \Rightarrow \sigma^2\delta(\tau) = \delta(\tau)$$

This was verified by applying the matalab function $xcorr(x,'unbiased')$ on WGN. As illustrated in figure(17) the ACF was symmetrical about $\tau = 0$,figure(18) and (19)

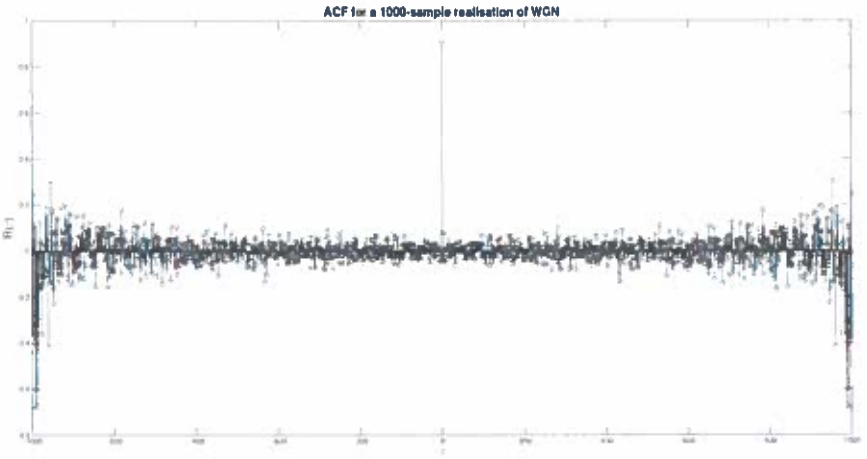
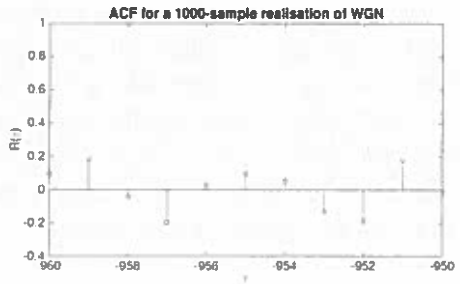


Figure 17: Ensemble statistics of rp3

Correct description of ACF and ACF estimate of WGN,

plot of zoomed ACF is missing



good caption

Figure 18: ACF of WGN for all $\tau[-999, -800]$

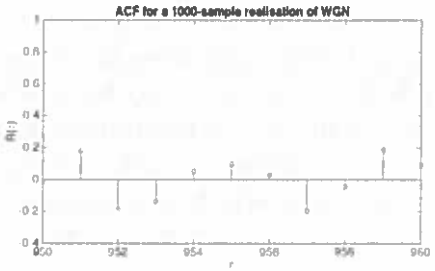


Figure 19: ACF of WGN for all $\tau[-1000, 1000]$ with references at ± 0.05

For small τ values in the range of $|\tau| < 50$ the amplitude of the ACF signal was small and for τ=0 the amplitude had a value of 1 as expected. For small delays it is therefore concluded that the samples appear uncorrelated and the estimator is indeed unbiased.

Correct explanation of reliability of ACF for large lags

On the other hand, for large τ values close to the limits, ACF appears to be considerable larger. This is occurring because as the lag increases, less data points are considered in the estimate of the ACF, increasing the variance. Taking into account the uncertainty for autocorrelation, it was observed that at large lag values th autocorrelation is proven to be unreliable. Ultimately a good empirical bound for the ACF is characterized by a $R_x(\tau)$ equal to zero everywhere apart from $\tau = 0$, therefore upon investigation it was found that the empirical boundary of τ is around $|\tau| \sim 150$.

2.2 ACF of correlated sequences

This exercise was focused on the application of a ninth order moving average(MA) filter on a 1000-sample WGN vector x. The shape and properties of the ACF were investigated to determine the relationship of each sample with the 9 previous or 9 next samples, and with itself.

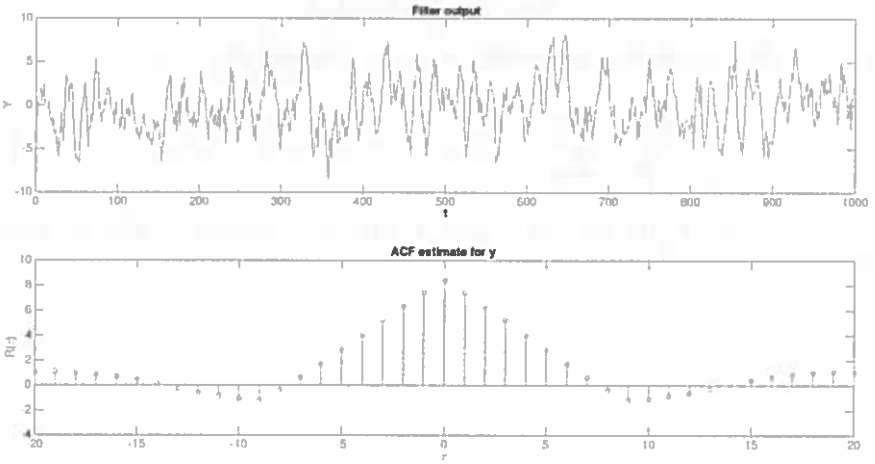


Figure 20: ACF of filter output

More explanations are required

As we can observe from figure (20),there were 9 spikes in each of the two intervals $-8to0$ and $0to8$, confirming the action of a ninth order filter. Longer delay values have ACF values which appear considerably lower as expected from the principles highlighted in $Part2.1$ Regarding the shape of the ACT of the filter impulse response, as the order increases the impulse response retains the same shape but it decays more slowly, in a way which is proportional to the order of the filter. Furthermore, the filter order also reflects the number of spikes that are produced at the output.

Good Mathematical proof

$$y_n = \frac{1}{q}\sum_{l=n}^{n+q-1} x[l] \Rightarrow y_1 = \frac{(x[1] + ... + x[9])}{9}, y_2 = \frac{(x[2] + ... + x[9])}{9}, ... \quad (4)$$

Looking at the above equation it was concluded that the first element of the series was equivalent to the mean. The rest of the elements cannot be considered as true averages, because as the series continues one less element is considered in the calculation each time.

2.The ACF of the stochastic process Y_n is given as :

$$R_Y(\tau) = R_X(\tau) * R_h(\tau) \quad (5)$$

where $R_Y(\tau)$ is the result of the convolution of $R_h(\tau)$, the ACF of the impulse response of the filter with $R_X(\tau)$,the input ACF of the signal. Since in this case X_n is an uncorrelated process, therefore $R_X(\tau)$ will become a dirac function with height equivalent of the variance of X_n at $\tau = 0$ and zero everywhere else. Therefore since the variance of X_n is 1, the convolution results in the following:

$$R_Y(\tau) = R_h(\tau) \quad (6)$$

2.3 Cross-correlation function

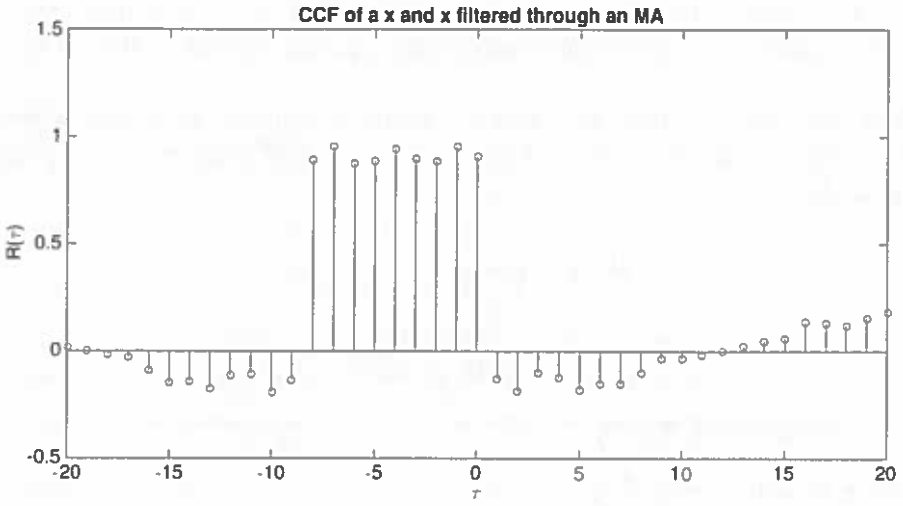
The cross-correlation function(CFF) is defined as the expectation of delayed samples of two different stochastic processes.

$$R_{XY}(\tau) = E\{X_n Y_s\} \quad (7)$$

The equivalent estimator of ergodic process is given by:

$$\hat{R}_{XY}(\tau) = \frac{1}{N - |\tau|}\sum_{n=0}^{N-|\tau|-1} x[n]y[n-\tau], \qquad \tau = -N+1, \ldots, N-1 \quad (8)$$

During this exercise sequences x and y will be considered and the estimate of their CCF will be plotted.



Correct plot of CCF of MA process

Figure 21: Cross-Correlated function of WGN and Y

1. Using matlab command $xcorr(x, y, 'unbiased')$ figure (21) was obtained. The shape of the CCF is as expected, with nine spikes having values of approximately 1. The rest of the function appears to be relatively flat with points close to zero. This is expected since the only correlation between the sequence x and y is between each sample, itself and the 8 past samples. That is the reason behind the spikes appearing between $\tau = 0$ and $\tau = -8$.

Looking at the CCF R_XY between the input and the output of the filter:

$$R_XY = h(\tau) * R_X(\tau) \tag{9}$$

The equation supports the conclusion because as mentioned before if X_n is an uncorrelated stochastic process then its ACF will be a dirac function. The resulting convolution yields that the CFF R_{XY} will have the same values as the filter's impulse response, with the only difference being that, due to properties of convolution, it will be mirrored about the y-axis.

Correct answer on how result is used for system ID

2.These results could be potentially be used for identification of an unknown system's response, by inputting a completely random signal and calculating the cross correlation of the output and the input. The system will be identified because the output will be the impulse response of the system in focus. Further improvements to the identification can be made by inputting a larger input signal because by increasing the number of samples comprising the input signal, the accuracy is improved. Having a filter of greater order causes the CCF to not give an accurate portrayal of the impulse response. The fact that higher order filters produce more non-zero elements at non-zero lag, affect the desirable results in a negative way.

2.4 Autoregressive modelling

The autoregressive modelling function of order 2 is given by:

$$x[n] = a_1 x[n-1] + a_2 x[n2p] + w[n]. \tag{10}$$

In this section the AR(2) is investigated with $a1$ being a random variable where $a_1 \in [-2.5, 2.5]$ and $a2$ is a random variable where $a_2 \in [-1.5, 1.5]$. The convergence

condition was checked by identifying whether the $|x(n)|$ was lower than 100. Using matlab the values of the pairs that result into x converging were plotted in a star graph.

Taking into account that the stability condition that the roots of a polynomial must lie inside a unit circle for stability to be true and knowing the z-transform equation of AR(2):

$$H(z) = \frac{1}{1 - a_1 z^{-1} - a_2 z^{-1}} \tag{11}$$

$$z_1, z_2 = \frac{a_1 \pm \sqrt{a_1^2 + 4a_2}}{2} < |1| \tag{12}$$

we can deduce the following:

- When $a_1 = 0$ then $a_2 = \pm 1$
- When $a_2 = 0$ then $a_1 = \pm 1$
- When $a_1^2 - 4a_2 = 0$ then $a_1, a_2 = (\pm 2, \mp 1)$

The conditions on the coefficients are therefore:

- $a_1 + a_2 < 1$
- $a_2 - a_1 = 1$
- $-1 < a_2 < 1$

Relatively Good plot of the Stability Triangle

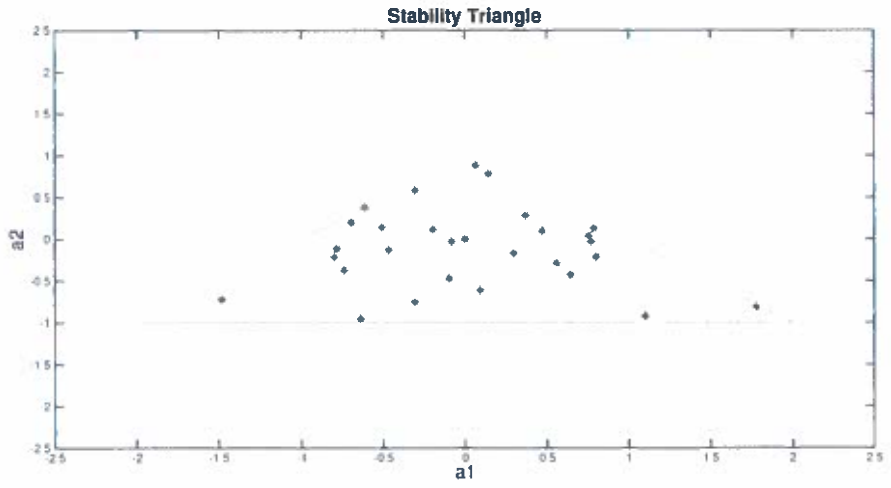


Figure 22: Stability triangle for AR(2) processes

As it can be observed from figure(22), the stability triangle was obtained. The area inside the triangle reflects the are where the coefficient pairs are stable. In addition the triangle can be split into 4 subregions.
Region 1: $a_2 > 0$ and $a_1 > 0$,defined a monotonically decaying ACF.
Region 2: $a_2 > 0$ and $a_1 < 0$,defined a decaying oscillating ACF.

Region 3: $a_2 < 0$ and $a_1 < 0$, defined an oscillating pseudo-periodic ACF.
Region 4: $a_2 < 0$ and $a_1 > 0$, defined a pseudo-periodic ACF.

2. This part of the exercise takes into account the real world sunspot time series. The time series is analyzed using *matlab* in order to investigate the ACF of the sunspot series for data lengths $N = 5, 20$ *and* 250.

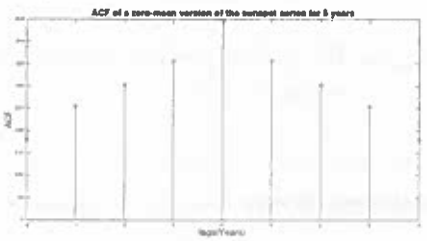


Figure 23: ACF of zero-mean sunspot time series for N=5

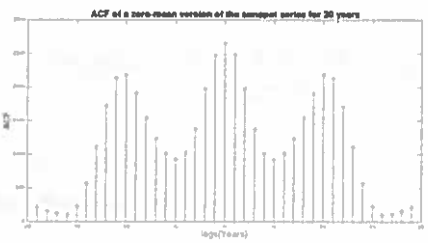


Figure 24: ACF of zero-mean sunspot time series for N=20

Figure 25: ACF of sunspot time series for N data length

Correct plots of ACF for Sunspot

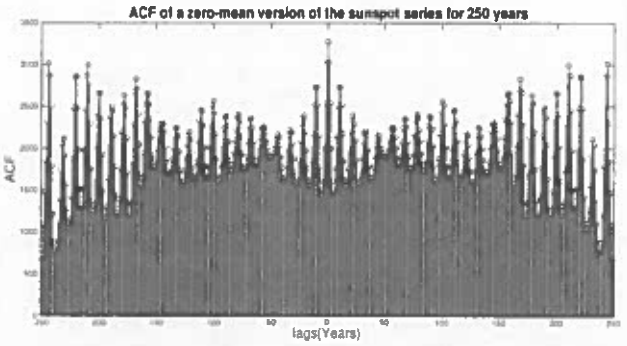


Figure 26: ACF of zero-mean sunspot time series for N=250

As illustrated in figure(23) for a relatively small number of samples a triangular profile is observed. As the number of samples increases, oscillations can be observed with decreasing ACF amplitude which never goes below zero. With very large sample numbers however, the ACF oscillations behave differently due to the large mean(48.4) which interferes with the changes in the sunspot values. Moreover, Figure(26) highlight the fact that despite not being able to observe correlation between the samples of the 5 year period, when 11-12 years are taken into account they are strongly correlated.

Good comment on the differences

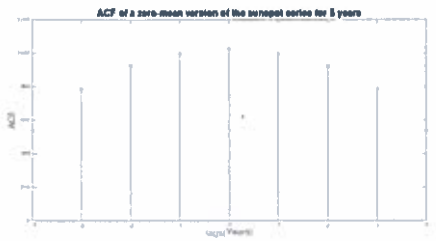


Figure 27: ACF of zero-mean sunspot time series for N=5

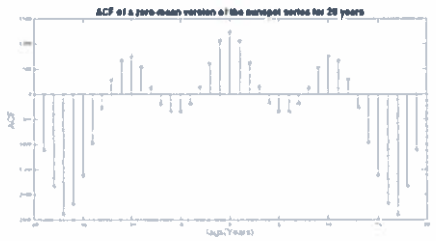


Figure 28: ACF of zero-mean sunspot time series for N=20

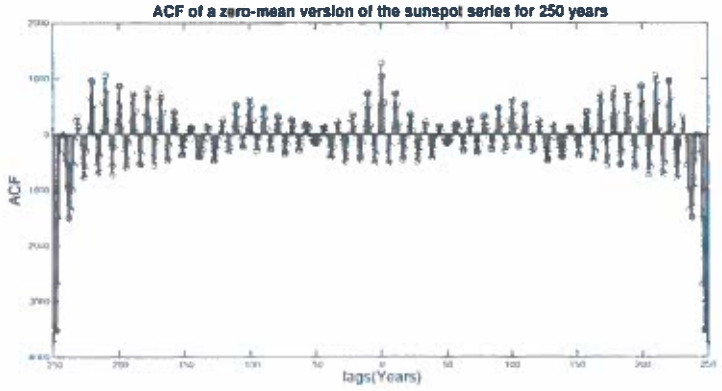


Figure 29: ACF of zero-mean sunspot time series for N=250

When the zero-mean sunspot time series is plotted, the shape of the resulting graph for $N = 5$ is the same, with a higher center-point,figure(27). The resulting ACF maintains the decreasing amplitude characteristic but it is shifts between positive and negative ACF values. Negative ACF values around 5 years apart, indicate no correlation. Positive ACF values around 12 years apart indicate strong correlation. The technique of removing the mean therefore assists us in obtaining a more realistic image of the autocorrelation function. This enables us to draw more meaningful conclusions by removing details that introduce distortion.

The Yule-Walker equations up to model order 10 are introduced in this part of the question using the *matlabcommandaryule* in an attempt to predict the most suitable order that models the sunspot time series depicted in figure(30) below. In order to obtain an accurate way of choosing the most suitable model, the partial autocorrelation function(PACF) *parcorr* was used.

Correct Identification of model order

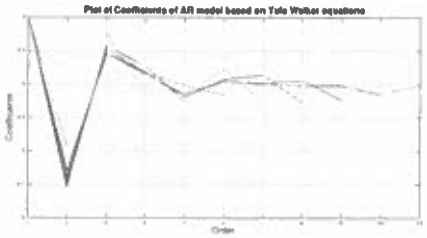


Figure 30: Yule-Walker coefficients for AR(p)for increasing order

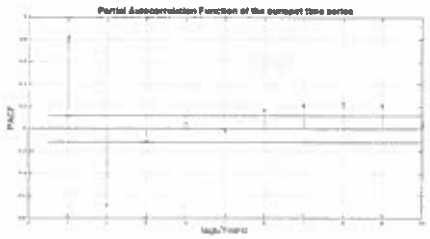


Figure 31: PACF of the sunspot series

Figure 32: ACF of sunspot time series for N data length

As we can see from figure(31), the partial autocorrelation function also draws a 95% confidence interval which helps us identify the most suitable order. In this case its clear that the most precise model is AR(2) with coefficients $a = [-1.47, 0.58]$. The same procedure was applied to the zero-mean, unit variance sunspot series. The difference whilst being compared to the previous results is that the coefficient are slightly smaller.The PACF proved to be the same as before further strengthening the conclusion that the AR(2) model is the most suitable one.
4. Aiming for the best suitable model for the sunspot series, the minimum description length(MDL) and the Akaike information criterion (AIC) were taken into account to find the optimum point characterising the smallest possible trade off between order and accuracy.

$$MDL = \log(E_p) + \frac{p\log(N)}{N} \tag{13}$$

$$AIC = \log(E_p) + \frac{2p}{N} \tag{14}$$

where E_p is the loss function, p is the model order and N is the number of estimated data points. It is expected that the optimum model order will minimise the MDL or AIC functions. The first step towards finding AIC and MDL is to find the cumulative square error(E),defined as the difference between the estimate and the real data.

$$E = \sum_{n=1}^{N} (x[n] - \hat{x[n]})^2 \tag{15}$$

Matlab was used to implement the error function to calculate the standardised MDL and the AIC.

Explanation of using AIC and MDL and how each terms effects the value is not complete

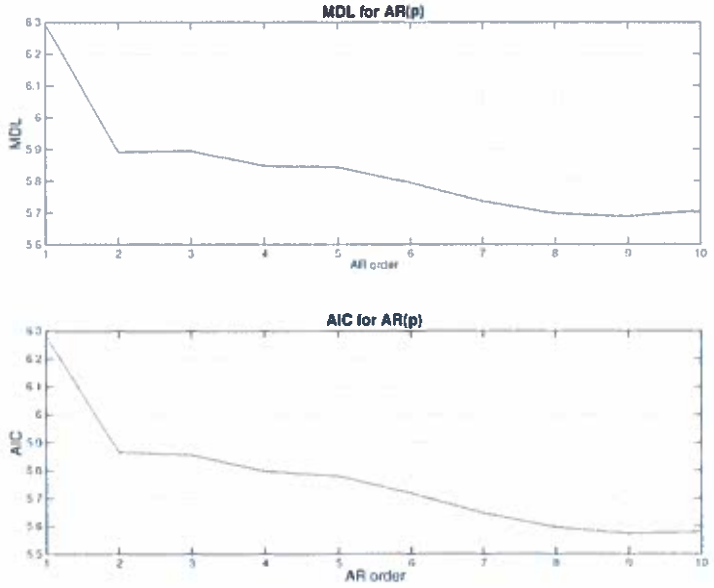
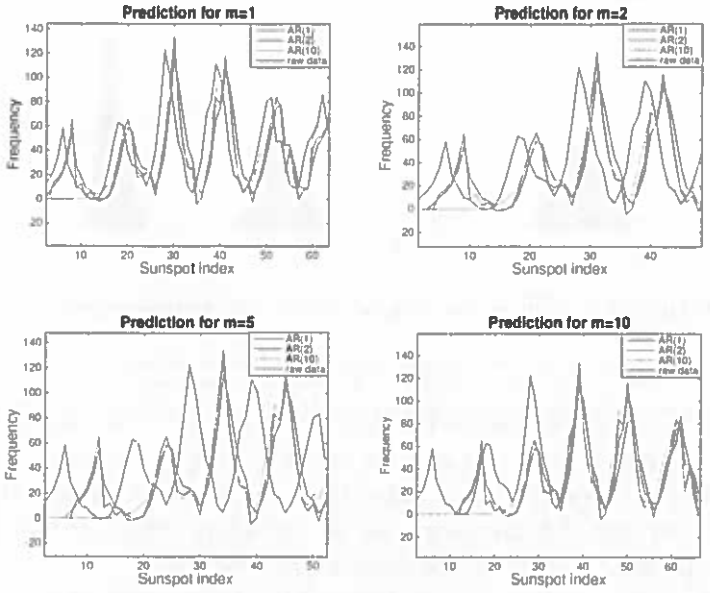


Figure 33: Plots of the MDL and AIC for a given AR order

By looking at figure (33) above we can clearly observe that the minimum value of both AIC and MDL occurs when the model order is 2. This leads to the conclusion that an AR(2) will give the best estimate for the sunspot series. 5.At this point the AR model described in the equation below is used to predict the sunspot time series m steps ahead.

$$\hat{x}[n+m] = a_1 x[n-1] + \ldots + a_p x[n-p]. \tag{16}$$

The analysis is originated from the 11th sample because since a model of order 10 is considered, the 10 past samples have to known. Figure (34) below illustrates the model's prediction horizon $m = 1, 2, 5, 10$.



Correct plots of predictions

Figure 34: AR(1) [blue], AR(2) [orange] and AR(10) [yellow] predictions of sunspot series [purple]

Eventhough as shown before, the optimal AR model corresponds to AR(2), the AR(10) has a profile which is very similar to the original. As it can be noted from figure(X) above, overmodelling applied to large prediction horizons can be more effective than a lower AR modelling order. However, caution must be taken with high AR modelling orders due to the increase in the error estimates, potentially leading to meaningless data. On the contrary, undermodelling has the advantage that the predictor takes in less data, decreasing the computational complexity. There exists the trade-off between prediction accuracy and computational complexity. Taking the aforementioned into account, it would be more appropriate to utilize a lower order model with a lower prediction error at small time step values.

2.5 Real world signals: ECG from iAmp experiment

Part 2.5 involves the use of real ECG data obtained in class corresponding to: 1.unconstrained breathing, 2.constrained breathing at 7.5bpm and 3. constrained breathing at 25bpm. Data from the unconstrained breathing were used in parallel with the following equation to calculate coefficients $h(0)$,$h(1)$ and $h(2)$.

$$h[n] = \frac{60}{rr[n]} \qquad \hat{h}[1] = \frac{1}{10}\sum_{i=0}^{10} \alpha h[i], \qquad \hat{h}[2] = \frac{1}{10}\sum_{i=11}^{20} \alpha h[i] \tag{17}$$

Correct plots

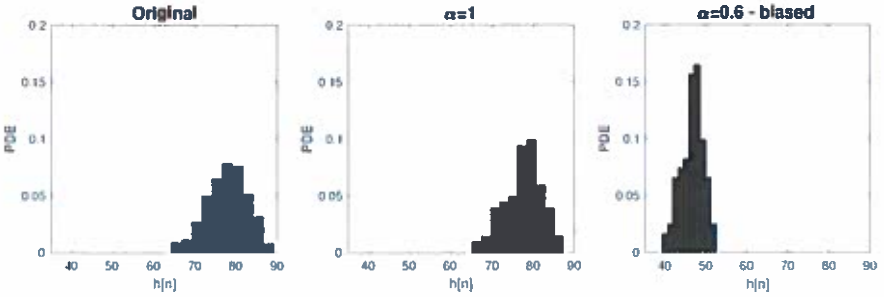


Figure 35: PDE of the original, a=1 and a=0.6 respectively

The three figures above show that the PDEs have relatively a very similar shape, however when the averaged method is applied, when a=1 the variance diplayed a decrease with a range between 67 and 86. upon setting a=0.6, a bigger decrease in variance was observed resulting in a range from 40-51. It can be said that by setting a=1 the estimator became unbiased, on the contrary when a=0.6 the estimator became biased.Biased estimators are often more desirable.
By taking a closer look at the plotted ACFs of the three trials, it was observed that for all the lags

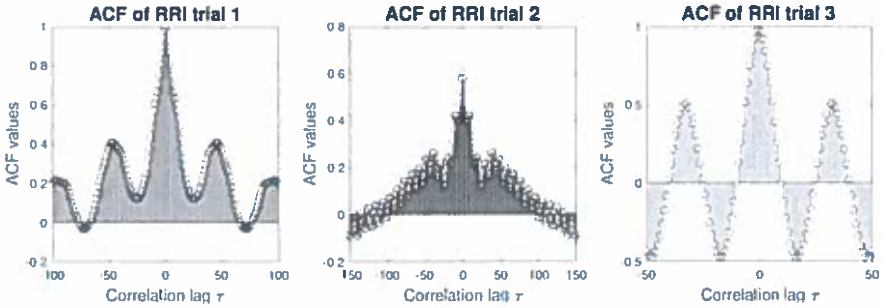
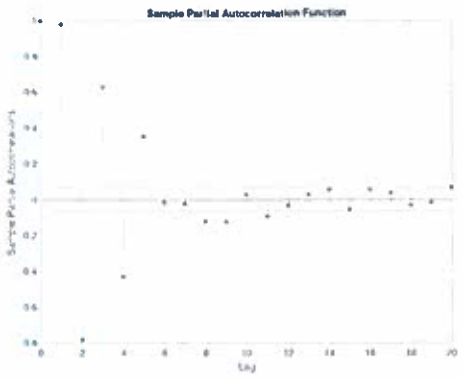


Figure 36: Autocorrelation of trial 1,2 and 3

By using the PACF as seen in figure (X), it was noted that the RRI data represent an AR model, whose optimal model order is 2.

Title and axis are small



The process of modal selection could've been more thorough

Figure 37: PACF of RRI data of trial 1

3 Spectral estimation and modeling

In order to further investigate ergodic processes the Power Spectral Density(PSD) was used. PSD is responsible for displaying how the power of the stochastic process X_n is distributed across its frequency components. This is characterized by the Wiener-Khintchine equation:

Good format of equations

$$P_X(f) = |\sum_{\tau=-\infty}^{\infty} x[n]exp^{-j2\pi f\tau}|, \qquad f \in [0,1] \tag{18}$$

In the case of real world applications ,an approximation of the infinite sum is used which is based on the FFT of the available samples. The outcome is called a periodogram.

$$\hat{P}_X(f) = \frac{1}{N}|\sum_{n=0}^{N-1} x[n]exp^{-j2\pi f\frac{n}{N}}|^2, \qquad f = 0, 1/N, \ldots, (N-1)/N \tag{19}$$

The processed above have unit variance as well as an ACF described by $\sigma^2 * \delta(t)$. In theory the PSD is expected to be a straight line at unity. On the contrary the PSD above display a poor estimate. This can be justified by the limited length of X_n.

3.1 Averaged periodogram estimates

This section focuses on mechanism that aim to smooth the PSD.
1. A zero-phase FIR filter with impulse response $0.2 * [11111]$ was used. The resulting periodograms were observed in figure(38).

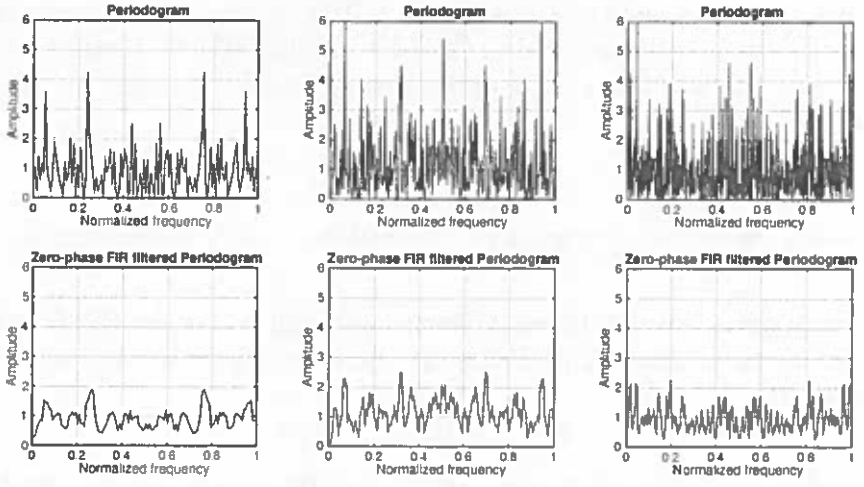
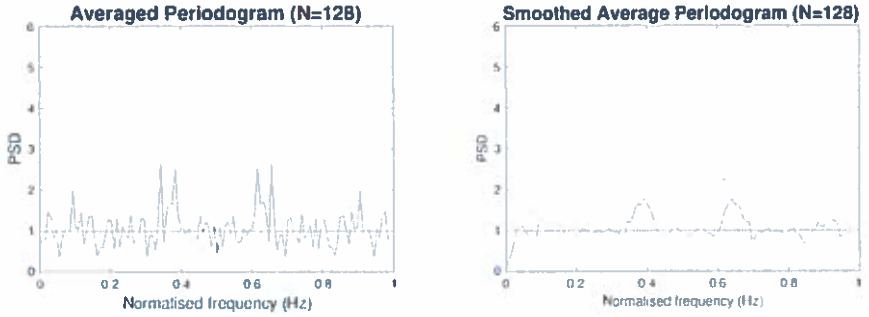


Figure 38: PGM of original N-sample realisation (top row) and filtered N-sample realisation of WGN(bottom row)

Having in mind that the desirable outcome is a horizontal line of PSD equal to unity, it can be seen from figure (38)'s lower row that the filter managed to remove a considerable amount of the spiky nature of the signal, proving that smoothing with

a MA is improving the PSD estimate.
2.For the second part of the question,a 1024-sample sequence of WGN was subdivided it into eight non-overlapping 128-sample segments. Upon examining the variation of the segments, it was observed that the PSD of the segments remained spiky with even twice the theoretical value. It can be therefore concluded that they share a great resemblance with each other and the same statistical properties.
3.The 8 segments were averaged to obtain the *AveragedPeriodogram*. By performing this averaging it is expected to reduce the variance by 1/(number of independent averaged estimates) as well as maintain the estimator unbiased. Figure(39) below highlights that the averaged PSD produced the smoothest results so far. As observed it displayed the a variance which was lower than the individual segments by a one order. Moreover, the technique of part 1 was used to further filter the averaged periodogram. This produced an even smoother outcome which concludes that the performance of the PSD can be improved with simple techniques of averaging.

Plot of the several period grams of the 8 segments are missing



Correct plot of the averaged periodogram

Figure 39: Comparison of averaged and averaged smoothed periodograms

1.A 1064-sample WGN sequence x was created using the MATLAB command *filter* with the coefficients $b = 1$ and $a = [1, 0.9]$, producing a filtered output y. The first 40 samples were removed to prevent transient effects. Taking into account the PSD of a random input with unit variance filtered through an AR(1) filter characterized by:

$$P_Y(f) = \frac{\sigma_X^2}{|1 + \sum_{k=1}^{p} a_k exp^{-j2\pi kf}|^2} \quad \Rightarrow \text{AR(1)} \quad P_Y(f) = \frac{1}{|1 + 0.9exp^{-j2\pi f}|^2} \tag{20}$$

In continuation, *matlab* function *freqz* was used to output the frequency response of the digital filter $H(z)$ Using the equation below the PSD of the output was calculated.

$$P_Y(f) = |H(f)|^2 P_X(f) \tag{21}$$

Looking at figure(40) below it is evident that the output displays smoothing of the input signal mimicking the behavior of a high pass filter. The cut-off frequency appears to be approximately 0.45rad/sample. Below the cut-off frequency, the input appears to be very small. In oppose, after 0.45rad/sample there is a gradual steepness,characteristic of a small order filter, allowing higher frequencies to pass.

Good Explanation of high pass nature

Good plot of AR spectrum

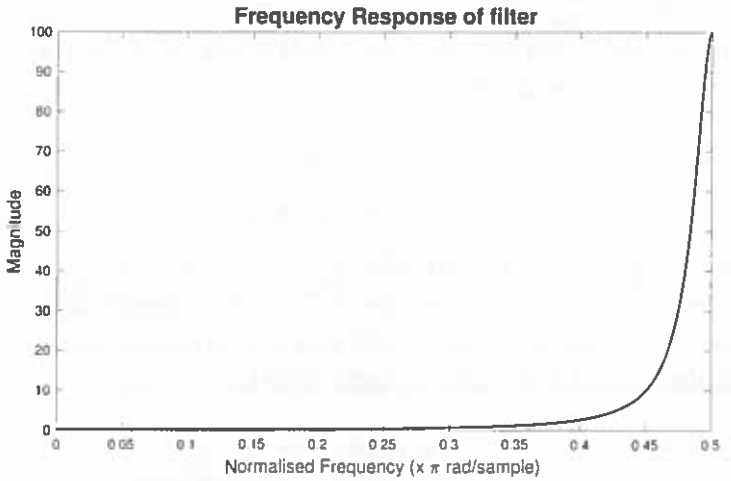


Figure 40: Frequency response estimation or AR1

2.This part focuses on the comparison of the previous outcome with the periodogram of y.

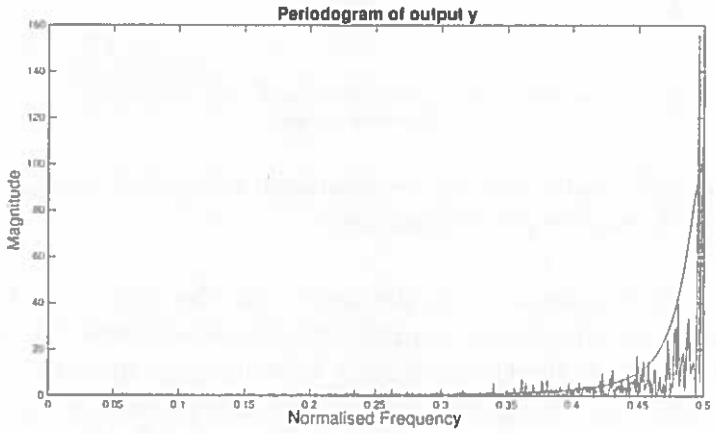


Figure 41: Theoretical(blue) and Estimated(red) PSD

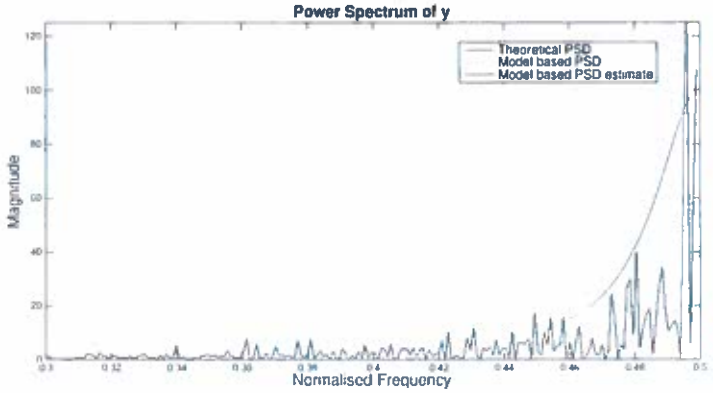
From figure(41) it was observed that the periodogram proved to be a poor estimate of $P_y(f)$. There is a resemblance between the two however there exists great variance between the theoretical filter response and the periodogram.
3.The close up in the interval [0.40.5] displayed that when low frequencies are considered, the periodogram and the theoretical are similar. As frequency increases towards the Nyquist frequency, the two become quite different with the introduction of large spikes in the case of the periodogram. This is occurring due to the fact that the data are discrete and of a certain size. Taking into consideration the rectangular window the periodogram estimator, a rectangular window of unity amplitude is used to get the discrete samples in our data. In the frequency domain the the rectangular window represent a sinc function. As a result, multiplying in the time domain with the rectangular window corresponds to performing convolution in the frequency domain of the sinc function with the spectral component of the data. The result of this convolution is the large oscillating variations observed in figure(41).

4. Model-based PSD is considered an other method of estimating the PSD. In order to compute the model-based PSD, the correlation function of the filter's output and parameters (a_1, σ_X^2) were used, where:

$$\hat{a}_1 = -\hat{R}_Y(1)/\hat{R}_Y(0) \tag{22}$$

$$\hat{\sigma}_X^2 = \hat{R}_Y(0) + \hat{a}_1 \hat{R}_Y(1) \tag{23}$$

(a_1, σ_X^2) was found to be 0.9243,0.669 respectively. In continuation, the *Matlab* function *xcorr* was used to calculate the ACF of the output signal of the filter. Figure(42) below illustrate the output $\hat{P}_y(f)$ which is equivalent to the PSD estimate, created by the theoretical PSD and the pgm function.



Correct estimation of AR coefficient and variance

Figure 42: PSD estimates showing the theoretical (blue) filter response, the model based PSD (black) and the periodogram (red)

It can be clearly seen from the figures above that the model-based PSD displays a superior estimate to the periodogram and the theoretical PSD. By taking a closer look at the profile of the model-based PSD, it is observed that there are no spikes with large variation and it appears smooth throughout. However, before making any final conclusions, the model-based approach requires further analysis to make a verdict on the optimal model order. In oppose while using the periodogram these requirements do not need to be taken into account, allowing direct estimation of the PSD from the output.

5.The above procedure was repeated for the sunspot time series. For the calculation of the model-based PSD *matlab* function *pyulear* was used. PSD was estimated for orders 1-10 in an effort to find the optimal order. Figure(43-46) below shows PSD estimates for increasing order 2,6,8,13 for sunspot time series. It was seen that order 8,depicted in figure(45), was the optimal one since it captured the desired detail. The following higher orders didn't show any significant difference in modeling the PSD, thus they could be avoided. For lower orders, despite having smooth plots for the model-based and the theoretical PSD, their small degree of freedom made them inaccurate.

Good AR spectrum for multiple AR models but the title and legend are small

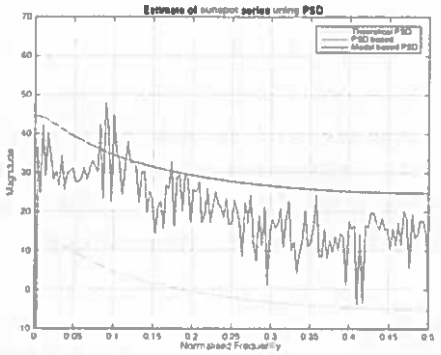


Figure 43: PSD of sunspot data order=2

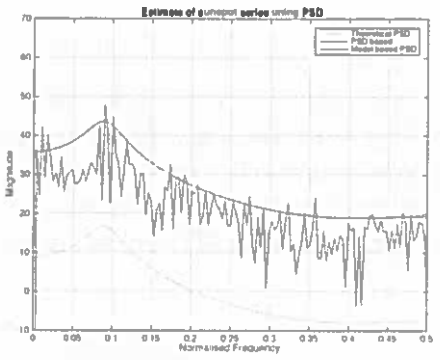


Figure 44: PSD of sunspot data order=6

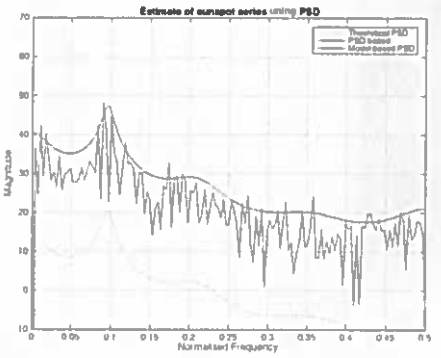


Figure 45: PSD of sunspot data order=8

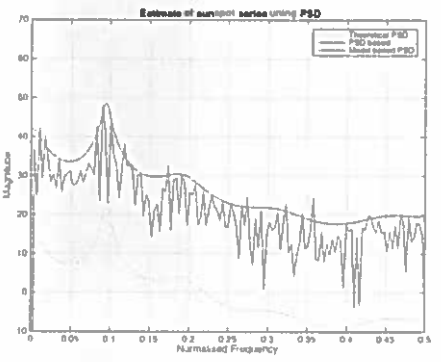


Figure 46: PSD of sunspot data order=13

3.2 Spectrogram for time-frequency analysis: dial tone pad

Time-frequency analysis through the use of the finite Fourrier transform is widely used in the sector of mobile telephone, the Internet and many more. This exercises focuses on the Dual-Tone Multi-frequency(DTMF) system consisting of two sinusoids:

$$y[n] = \sin(2\pi f_1 n) + \sin(2\pi f_2 n), \tag{24}$$

where f_1 and f_2 are the frequencies in Hz, and n is the time index. Table (3) displays the frequency pairs for each digit.

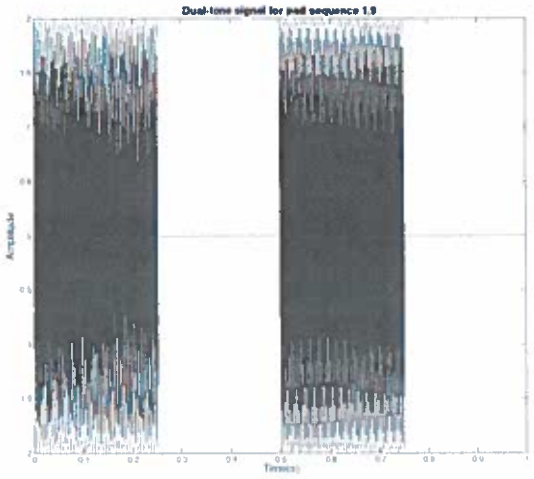
Table 3: Dial pad frequencies

	1209 Hz	1336 Hz	1477Hz
697 Hz	**1**	**2**	**3**
770 Hz	**4**	**5**	**6**
852 Hz	**7**	**8**	**9**
941 Hz	*	**0**	#

1. The *Matlab* function *randi* was used to generate a random UK land-line number:

$$0-2-0-6-6-2-8-1-8-3-2 \tag{25}$$

A sampling rate of $32768Hz$ was used to generate the discrete time sequence for each key pad. Each key pad was assumed to be pressed for 0.25[s]. 8192 samples are present in the 0.25[s]$(0.25 * 32768)$. Upon concatenating the idle sequences with the keypads of choice, the discrete time sequences of the UK landline number was constructed. Figure (47) below shows the generated tones for keypad-1 and-9.



time domain plot of the phone number is not complete

Figure 47: Signal y for key 1 (left) and key 9 (right)

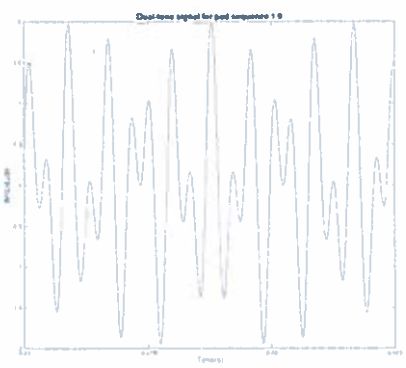


Figure 48: Two-tone signal for the key 1 withing the DTMF system

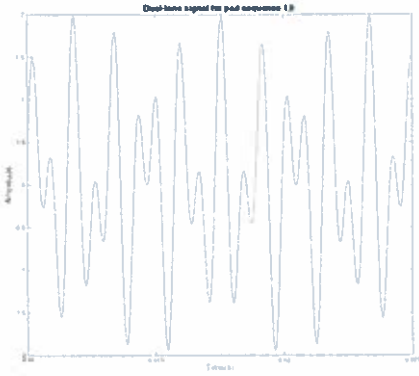
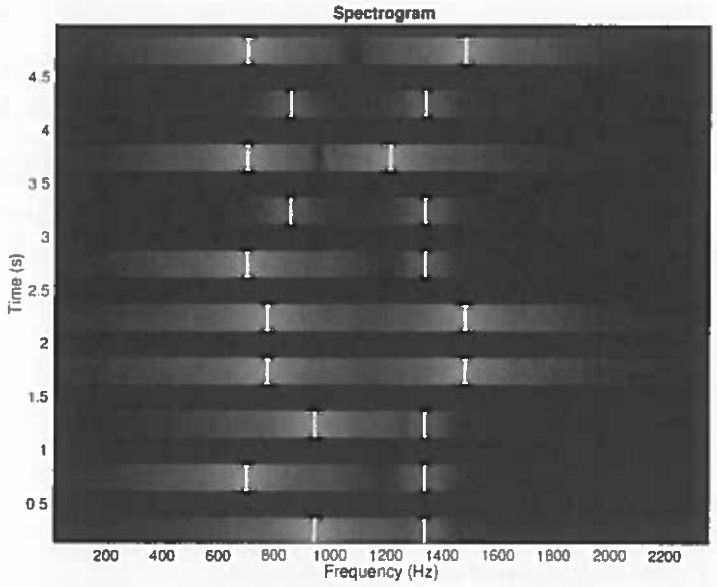


Figure 49: Two-tone signal for the key 9 withing the DTMF system

Upon adjusting the zoom and the axis parameters, it was observed that the two tones have different profiles. However, they share some common characteristics like symmetry and periodicity. Furthermore judging from the nature of the oscillating picks, it can be said that is comprised from 2 sinusoids, as expected.
2.The *matlab* function *Spectrogram* was used to analyse the spectral components of y. Care was taken while handling the function's parameters, as it important

not to include overlapping segments with FFT calculations. The syntax used was the following: $spectrogram(x, window, noverlap, f, fs)$,where x=The landline sequence y;$window$=Hanning window(8192, as each ton has this many number of samples);$noverlap$=no samples of overlap(0,for correct tone separation);f=cyclical frequency(8192,all samples are needed in this case);fs=sampling frequency(32768HZ). Figure(X) displays the resulting spectrogram.



Relatively good spectrogram

Figure 50: Spectrogram of the landline number

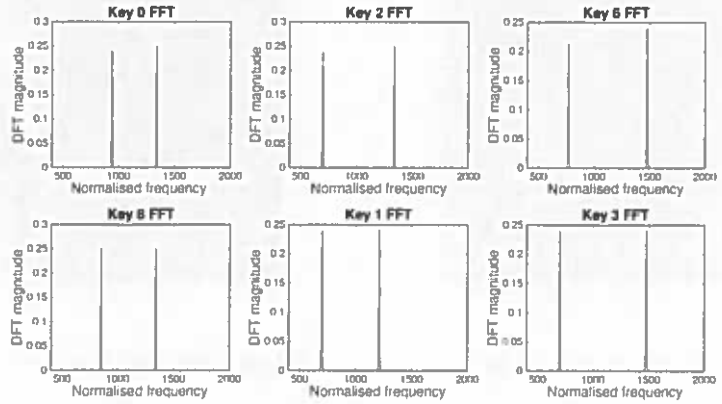


Figure 51: FFT of individual tones

As expected, it can be observed that every 0.25s a number is dialed. The frequencies of the dialed tones are depicted in yellow and the dark blue background corresponds to the idle tone which is characterized by virtually zero power. Furthermore, the gradual dimming seen after the bright yellow segments occurs as a result of the hanning window. Figure(50) shows a breakdown of the spectrogram as it displays the $|FFT(f)|$ of each of the tone segments. The digit identification is also displayed more clearly in the table(X).

	Spectrogram Output										
Intervals(s)	0.25	0.75	1.25	1.75	2.25	2.75	3.25	3.75	4.25	4.75	5.25
f(1)	950	700	950	770	770	700	850	699	850	700	700
f(2)	1340	1340	1340	1480	1480	1340	1340	1220	1340	1480	1340
Keys	0	2	0	6	6	2	8	1	8	3	2

Table 4: Frequncy ranges obtained from spectrogram

Explanation of key classification using spectrogram and table is not complete

3. In order to perform the identification of sequence y, table(3) and the spectrogram were used in parallel. The frequency pairs within a specific range were matched with the corresponding values from table(4). An example of this procedure is the time interval[0.50.75] with frequencies 700 and 1340, corresponding to key-2.
4. In the real world most signal get corrupted by channel noise. To investigate this conditions, white noise was added to the sequence. Three white noise cases were tested below.

1. Noisy signal with low variance=0.01
2. Noisy signal with medium variance=10
3. Noisy signal with high variance=100

Figures(52),(53),(54) below illustrate the three cases.

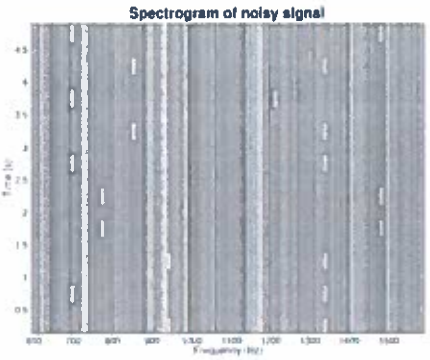


Figure 52: Spectogram for Var=100

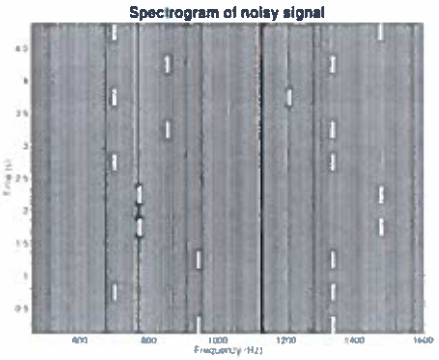


Figure 53: Spectogram for Var=10

Different spectrogram plots for various noise levels are presented

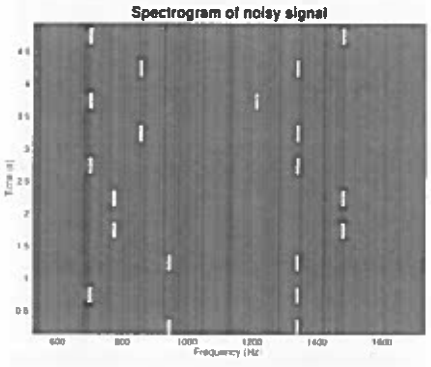


Figure 54: Spectogram for Var=0.05

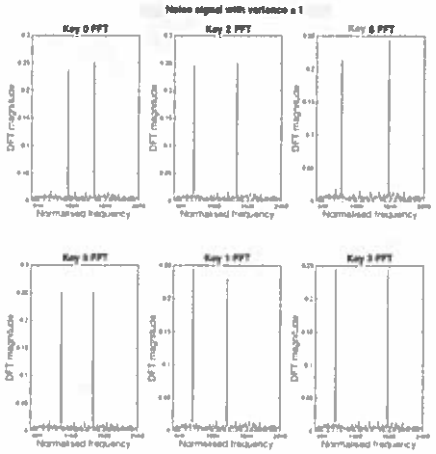


Figure 55: FFT of individual noisy tones

For the first case it was observed that the signal was distinguishable despite the noise. Successful tone identification was therefore achieved, as the spectrogram displayed clear bands of the characteristic tone frequencies. In the second case, despite that the signal was hardly distinguishable being immersed in noise, the spectrogram had still successfully identified the characteristic tone frequencies. Lastly in the third case the signal was not distinguishable at all. The spectrogram still managed to identify the characteristic tone frequencies, however it was considerably less clear to point out the highlighted parts corresponding to high power signals.Furthermore it can be seen that as the variance increases above a particular point close to 1, it is not possible to distinguish anything from the time domain, not even the idle sequence. Ultimately,looking at the background of the spectrogram, the colour is changing from a blue to a green/yellow for high variances. From this we can note that the idle tone gets very easily corrupted, which could result in mistaking a part of it for a tone, when the variance is too high.

3.3 Real world signals: Respiratory sinus arrhythmia from RR-Intervals

a) The RRI data from section 2.5 were loaded. Raspitory sinus arrhytmia(RSA) refers to the modulation of cardiac function by respiratory effort. Recalling from the previous exercise, trial 1 corresponds to unconstrained breathing,trial 2 corresponds to 25 breaths per minute and trial 3 corresponds to 7.5 breaths per minute. The standard and the averaged periodograms were applied to the data, with different window lengths, to investigate the PSD of the RRI data.

Nice figures, but smoothness requirement is not fully satisfied

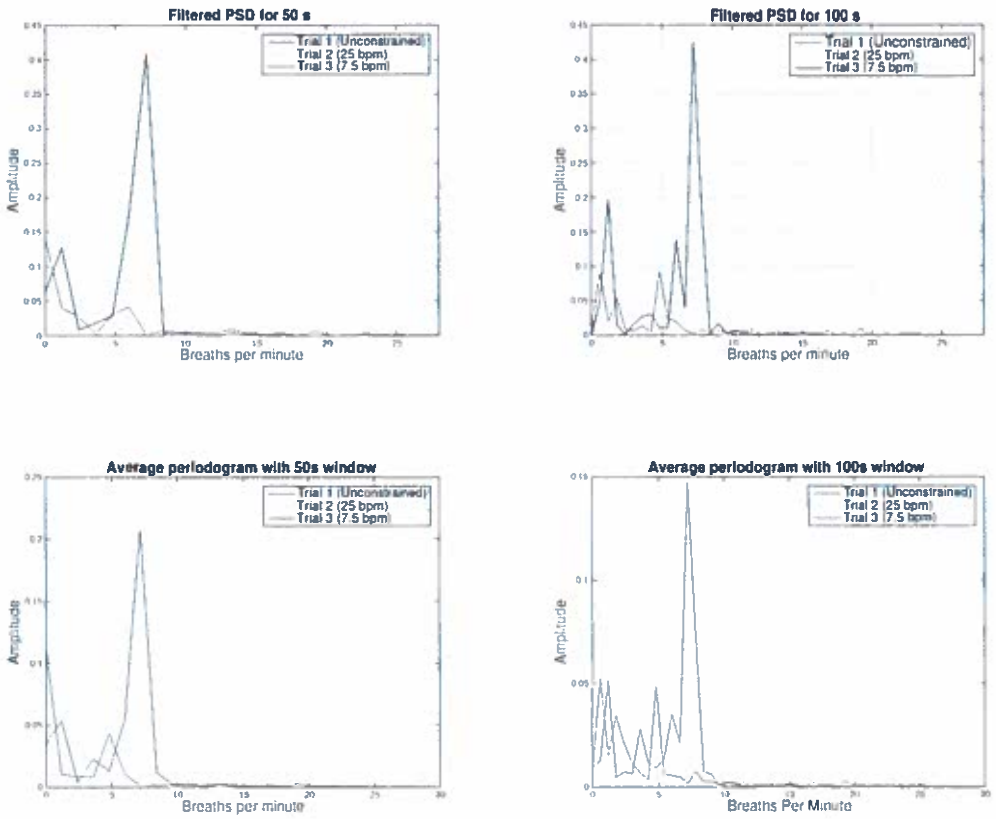


Figure 56: Filtered and averaged periodograms for the breathing data with windows 50s and 100s

The use of a larger window(100s) it was observed to affect the peacks by making them narrower. It can be seen that the PSD of each of the three trials enables us to identify the breathing rate that were imposed to us by the experimental procedure. As observed from the plots above in figure(56) the equivalent peaks can be matched with the unconstrained breathing of nearly 5 bpm, the constrained breathing at 7.5 bpm and the constrained breathing of 25 bpm

4 Optimal filtering - fixed and adaptive

4.1 Wiener filter

1. The first part of this exercise focuses on statistical variables R_{xx} and p_{zx} representing an autocorrelation and a cross correlation matrix are used to calculate the optimum Wiener filter solution:

Rxx and Pzx coefficients are missing

$$w_{opt} = R^{-}1_{xx}p_{zx} \tag{26}$$

A WGN sequence was generated acting as an input to the unknown system. The system was described by filter coefficients $a = [1]$ and $b = [12321]$. The filter's normalised output y_n was combined with the noise to generate z_n. Theoretically, the SNR is expected to be 20 given the equation below

$$\frac{P_y}{P_n} = \frac{\sigma_y^2}{\sigma_n^2} = \frac{1^2}{0.1^2} = 100 = 20dB \tag{27}$$

Subsequently the signal to noise ratio was calculated to be $19.9dB$, a value very close to the theoretical. As observed from the table below the optimum coefficients of the Wiener filter were relatively similar to the unknown system coefficients, reflecting accuracy, nonetheless small differences are present due to the small sample number an the noise. The same procedure was run for a range of selected variances of the additive noise where ($\sigma^2 \in [0.1, 10]$). The results were tabulated in table(5).

Correct SNR calculation and sufficient explanations

σ^2	0.1	0.9	1.8	3.6	7.2	10.0
w_{opt}	1.038	0.932	1.141	1.070	1.3134	1.007
	1.990	2.271	1.890	1.751	1.889	2.366
	3.064	2.902	3.262	2.867	3.555	2.181
	2.031	1.043	2.287	1.887	1.297	1.437
	1.058	-0.432	1.0347	1.169	1.401	1.520
$SNR_{dB}(1)$	10.28	0.49	-2.08	-5.80	-8.43	-9.702
$SNR_{dB}(2)$	10	0.46	-2.55	-5.56	-8.55	-10

Table 5: $\vec{w}_{opt}$, (1)Experimental and (2)Theoretical SNR for $\sigma \in [0.1, 10]$

Good complexity analysis

It can be observed that as the noise variance increases, the SNR is decreased causing the Wiener coefficients to become worse not permitting correct system identification. As N_w is increased it was noted that for $N_w > 4$ the coefficients increase but their value is very close to zero allowing them to be neglected. It can be therefore concluded that the optimal N_w is 4. Furthermore it must be said that the value of N_w does not affect the real estimated coefficient's accuracy. Taking into account computational complexity,it can be seen that the aforementioned algorithm is block-based requiring all the data before computing the optimal solution. The p_{zx},R_{xx} matrices need $N(2N1)$ multiplications and $(N1)(2N1)$ additions. Matrix w_{opt} needs $O(N_w^3)$ operations for the matrix inversion, N_w^2 multiplications and $N_w(N_w - 1)$ additions. The three matrices therefore involved and require a total of $(O(N_w^3) + 8N28N + 2 + 2N_w^2 N_w)$ operations resulting in a large drawback.

4.2 The least mean square (LMS) algorithm

The Wiener solutions is not able to identify non-stationary unknown systems since it assumes stationarity. This part of the exercise will focus on the use of adaptive filters. The least mean square algorithm defined by:

$$w(n+1) = w(n) + \mu e[n]x(n) \quad \text{what is this?} \tag{28}$$

is used as a simple basis for an adaptive filter, with $\mu = adaptationgaincontrollingthestabilityalgoritl$
A *Matlab* function *lms* was created to simulate the filter.
2. The least mean square(lms) algorithm was applied to the previous signals x and z_n, with initial adaptation gain=0.01. The evolution of the filter coefficients was plotted in figures(57-59).

Good Plot of the evolution of the coefficients, but legend is small

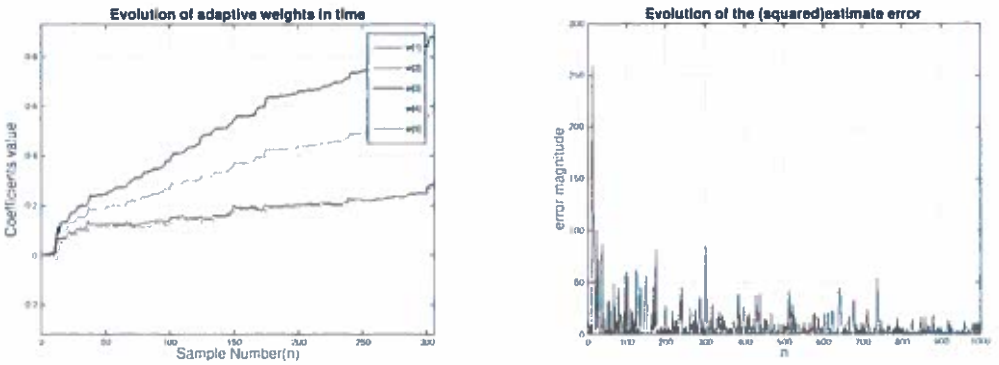


Figure 57: $\vec{w}_{opt}(n)$ and $e(n)^2$ for input signals with σ^2=0.01

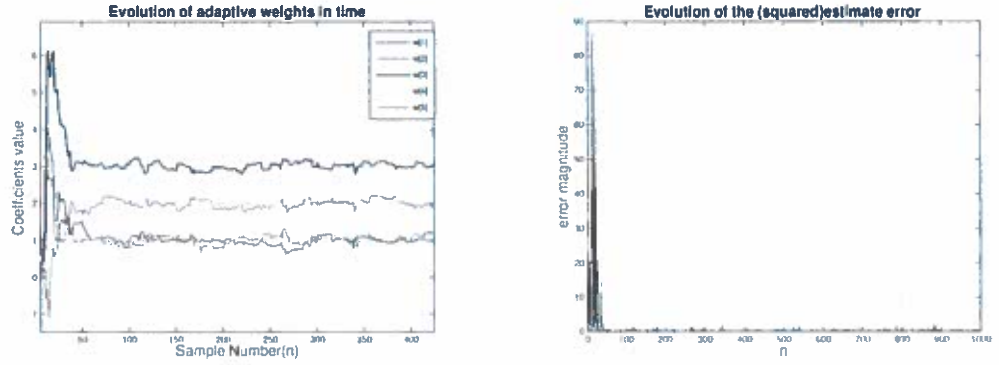


Figure 58: $\vec{w}_{opt}(n)$ and $e(n)^2$ for input signals with σ^2=0.1

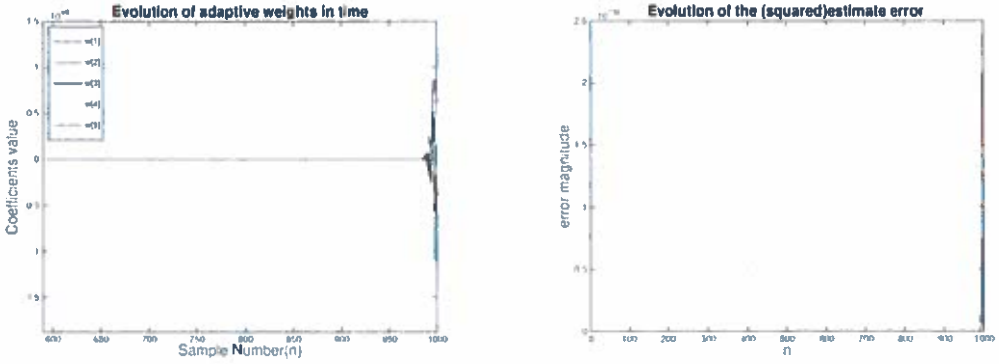


Figure 59: $\vec{w}_{opt}(n)$ and $e(n)^2$ for input signals with σ^2=0.7

As it can be observed from figure(58), the filter coefficient appear to converge to the Wiener solution values(1,2,3) examined in the previous part. The value of 0.01 is relatively small reflecting the small convergence of the coefficients that reach steady state at around 300 samplesIt was also observed that as the value of μ increases the convergence is minimised, but accounts for a higher error. Furthermore as the value of μ exceed a certain threshold around 0.3 the algorithm becomes unstable,with oscillations appearing on the coefficient values. In the case of even higher μ values

Complexity analysis is incomplete

the oscillations appear to big a lot bigger with no clues of reaching their value, but this can be explained by the divergence at high sample numbers.Regarding the complexity, $w(n+1) = w(n) + \mu e[n]x(n)$ requires $2(N_w + 1)$multiplications and $(N_w + 1)$ additions. $y[n]$ requires $(N_w + 1)$ multiplications and N_W additions and e_n requires one subtraction. This produces a total of $5N_w + 4$ operations.

4.3 Gear shifting

In this part of the exercise gear shifting was used as a method aiming to reduce the adaptation gain and improve the estimates at steady-state. The error is used in this case as a measure to decrease the value of the adaptation gain, resulting in a more accurate approximation. The *lms* function used before was modified to a new function lms_gs and it was used to find the optimal gain.

More gear shifting ideas could've been proposed

Legend is too small

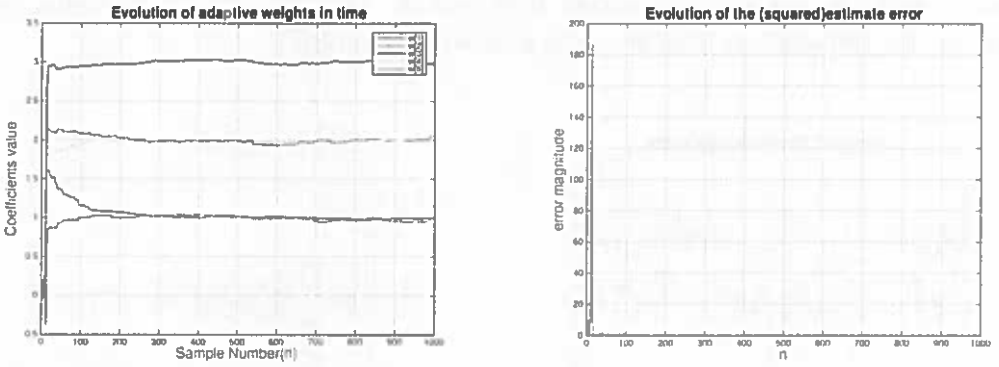


Figure 60: Time evolution of filter coefficients using gear shifting and corresponding error magnitude plot

As it is evident from figures (60), the gear shifting method produces clear enhancements. It as the fastest rising time so far as well as almost no oscillation present when the coefficients settle down to their true values. In terms of the error plot,figure(60-left), despite the initial overshoot,the steady state error appears negligible compared to the previous ones calculated through *lms*. It can be therefore concluded that a clear improvement is made both in terms of speed and accuracy without any major drawbacks.

4.4 Identification of AR processes

1.Speech recognition is amongst one of the many examples where adaptive filters may be used since it involves non-stationary signals which require adaptive estimation. A program was written in *Matlab* implementing a synthesis and analysis structure for an AR model, with parameters $a = [1, 0.9, 0.2]$. The values at which $a1$ and $a2$ should converge to are -0.2 and -0.9. The way the filter operates is by trying to obtain the initial signal by cancelling the effect of the input to the AR. It achieves this by having the exact negative coefficients at the output, to cancel each other out. Figure (61) below displays the coefficient values that appear to be converging to values -0.2 and -0.9.

Correct plot of the evolution of the AR coefficient

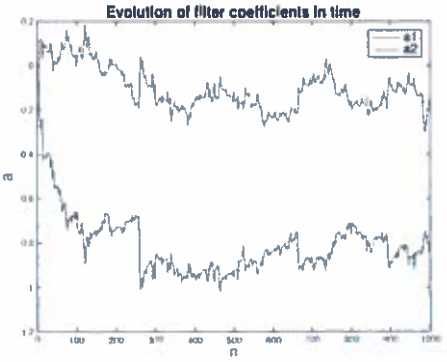
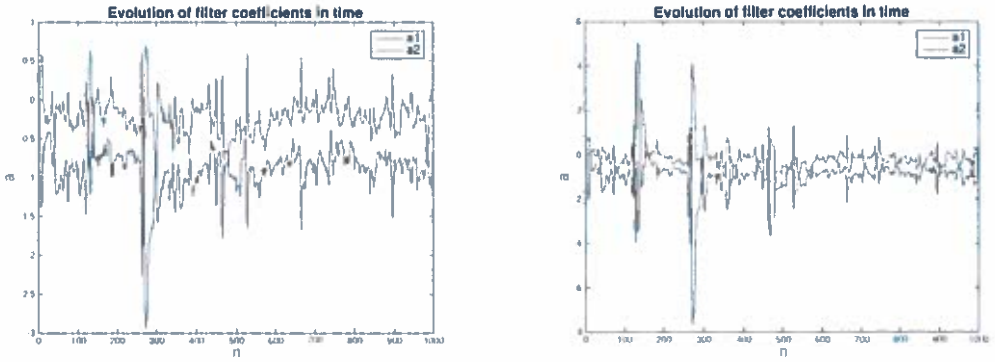


Figure 61: Time evolution of adaptive weights a1,a2

2. The adaptation gain was then varied in the interval $0.01 - 0.1$. The values chosen within the range are $\mu = [0.01, 0.05, 0.07, 1]$. The time evolution of filter coefficients for the selected μ values were plotted in figure(62).



Reasonable explaination of tradeoff between convergence speed and steady state error.

Figure 62: Time evolution of a[n] for m = 0.05 and 0.1

Looking at the resulting plots,as expected, it can be concluded that for small μ values(0.01) the system converges with longer settling times and smaller steady state error. As the adaptation gain increases the settling time decreases but more oscillations are present. When $\mu \geq 0.5$ the results appear unstable with distortion and high oscillatory activity.Concluding it can be set that the *lms* algorithm proved to be effective within a certain range, outside that error magnitude increases almost tenfold.

Plot of the time domain signal of speech is missing

4.5 Speech recognition

To test the speech recognition abilities of the filter, characteristic phonetic sounds were used as an input, and the filter is found the coefficients $a_1, a_2, ..., a_p$ in a way that the average output squared error is minimised. Having the input as a delayed version of the sound causes the estimator to output the sound itself,therefore being referred as a predictor.

1. The phonetic sounds of 'a' 'e' 's' 'x' and 't' were recorded. A sample 1000-long sample was used to capture the most representative range of frequencies for each

phonetic, at $f_s = 44100Hz$. Subsequently, the adaptation gain and order of the predictor were investigated to obtain the optimal values for each letter. First a range of values of adaptation gain was plotted again the prediction gain(Rp) to determine the optimum value of μ(x-value corresponding to the maximum point of the graph) for each letter separately. The same approach was carried out to find the optimum order for the predictor. A range of order values were examined and plotted against the predictor gain. The optimum order was selected by looking at the point where Rp was not showing any considerabl additional increase with increasing order. The results are depicted below in table(6).

	Model order	Adaptation gain	Prediction gain
letter 'a'	2	0.56	14.16
letter 'e'	8	0.32	13
letter 's'	12	0.15	6.88
letter 't'	24	0.008	2.89
letter 'x'	23	0.0.009	4.36

Table 6: Information regarding μ,order(p) and prediction gain(Rp) at fs=44.1kHzs

It can be observed that vowel phonetic sounds 'a' and 'e' display a higher prediction gain and appear more stationary. On the contrary, consonant phonetics 's','x' and 't' which require higher orders and do not appear very stationary. The most difficult phonetic was 't'. It is also suggested that gear shifting is used for higher prediction gains. It was noted that whilst in the previous part of the question there was a threshold value of about >0.4, in the case of phonetic sounds the threshold value is higher taking into account the fast-paced variation in speed even when coefficients appear stable. Gear shifting is discarded since large μ values are optimal in his case minimising the error.
2.Two approaches where examined to find the optimal length filter. Regarding a heuristic approach the table above can be used to locate the maximum Rp value. For the case of vowels it can be concluded that an order range between $3-8$ will be satisfactory. In the case of consonants the order range necessary to correctly map the non stationary signal is $13-18$.Regarding the analytical approach, the AIC and the MDL functions used in the previous exercise as well as the PACF were used. Resulting plots for the AIC and MDL are show in figure(63)

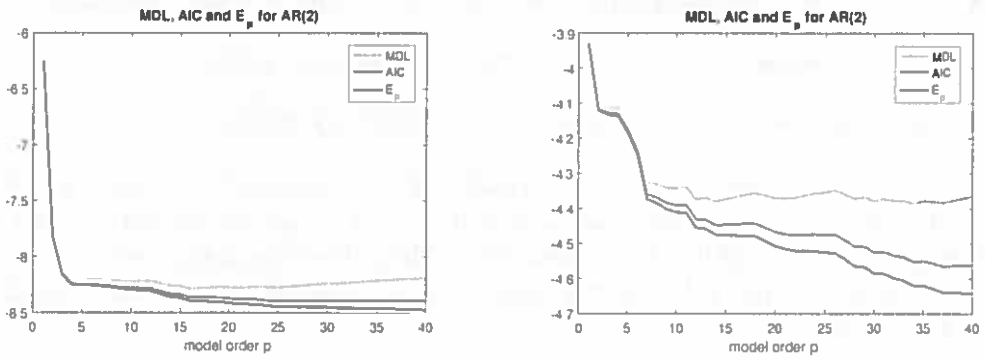


Figure 63: MDL, AIC and Ep plot for phonetic 'a' right and 't' left

This provides further information to obtain a clearer answer to whether increasing the order and thus increasing complexity is worth the resulting reduction in error. It was observed that an AR(3) appears to model accurately the signal for 'a'. In addition a higher AR(6) is needed to map the consonant 't' strengthening the difference in the ability for speech recognition when it comes to consonants.

3.For this part of the question the sampling frequency was changed to $f_s = 16000Hz$. The same procedure described above was repeated to find the optimal order and adaptation coefficient. The results are tabulated below in table(7):

plot of coefficients showing quasi-stationary weights is missing

	Model order	Adaptation gain	Prediction gain
letter 'a'	2	0.56	8.89
letter 'e'	8	0.32	8.83
letter 's'	12	0.15	1.42
letter 't'	24	0.008	1.31
letter 'x'	23	0.0.009	4.41

Table showing prediction gains is presented in the report

Table 7: Information regarding μ,order(p) and prediction gain(Rp) at fs=16kHz

Looking at the results of the different letters, despite the change in sampling frequency, the order of the filter remains relatively unchanged. The time between consecutive samples is calculated using the equation $t = N(samplenumber)/f_S$ to be 0.02268 and 0.0625 for the case of $44100Hz$ and $16000HZ$ respectively. It can be noted that the increased time between signals leads to lack of the desirable convergence due to the increase in the estimate error,forcing the coefficient to adapt to the ideal value more gradually and with a greater oscillatory behavior. This results in potential loss of information and a worst predictions performance. It can be therefore concluded that if a smaller sampling frequency is used it should be accompanied by either a large amount of samples or to reduce the adaptation gain to increase stability.

4.6 Dealing with computational complexity: sign algorithms

The sign LMS algorithms are given by:

- signed − error (sign algorithm) $\Rightarrow$ w(n + 1) = w(n) + μsign (e[n])x(n)
- signed − regressor $\Rightarrow$ w(n + 1) = w(n) + μe[n]sign (x(n))
- sign − sign $\Rightarrow$ w(n + 1) = w(n) + μsign (e[n])sign(x(n))

The performance of each algorithm is compared with the basic *lms* function used before for the AR parameter identification in Part 4.4, and for phonetic sound 'a' in Part 4.5. Figure (64-66) below illustrates the difference performances for the phonetic sound 'a' and table (8) the differences between the normal and the gear shifted expressed in Rp.

Algorithm Type	Prediction gain(Rp)
normal	7.56
gear shifted	7.64
sign-sign	3.31
sign-regressor	3.42
sign-error	3.39

Table 8: Information regarding μ,order(p) and prediction gain(Rp) at fs=16kHz

Good plots for different algorithms, but title, legend and axis are small. Also the space could've been used better

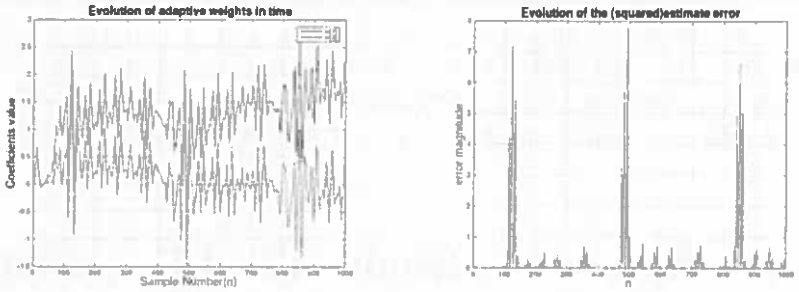


Figure 64: Time evolution of coefficients and equivalent error plot for sign-error algorithm of phonetic 'a'

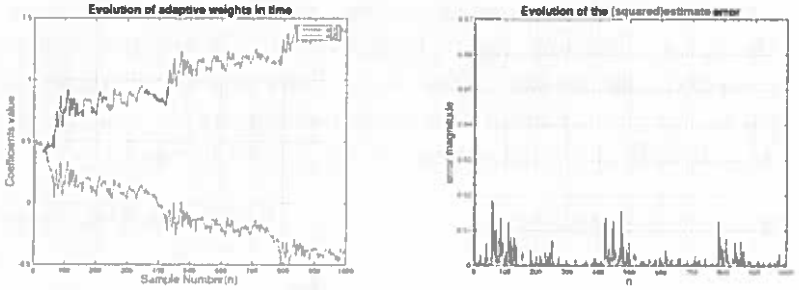


Figure 65: Time evolution of coefficients and equivalent error plot for sign-regressor algorithm of phonetic 'a'

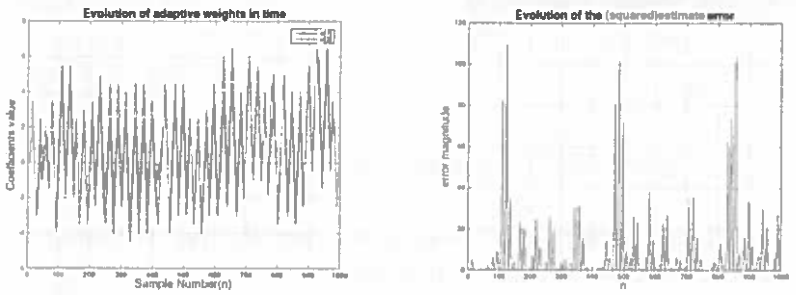


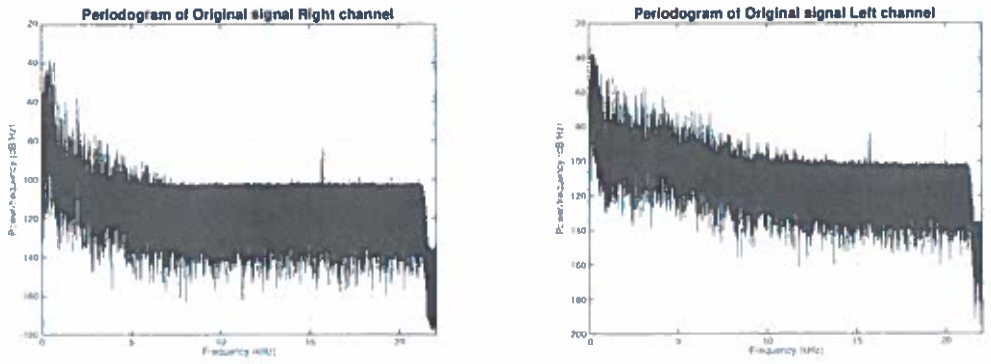
Figure 66: Time evolution of coefficients and equivalent error plot for sign-sign algorithm of phonetic 'a'

It can be observed that for the case of sign-sign algorithm there is an imporvement with the estimate as the sample number increases. For the case of the signed-regresor, the algorithm converges slower than the standard *lms*, but the error involved is smaller. Subsequently, the three algorithms were tested with the phonetic sound 'a',with $\mu = 0.56$ and order=5. As it can be seen from figure(66), the sign-regressor and sign error achieve a faster convergence compared to the lms, however resulting into different converging values. Upon looking at the error for the three algorithms, the sign-regressor provides the best results. This can be explained by the fact that the signal gets overestimated. Non-stationary signals tend to exhibit quick quick variations which inevitably require more adaptation. Following the same line of logic the sign-sign algorithm produces the biggest error due to its oversimplification property. ultimately, taking into perspective the complexities of the algorithms we can note that the sign-operator reduces the multiplications. This lead to the sign-sign algorithm needing $(N_w + 1)$ multiplications and $2N_w + 1$ additions. The sign-regressor and sign-error requiring $s(N_w + 1)$ multiplications and $2N_w + 1$ additions.

complexity analysis is presented in the report

5 A Real World Case Study: Vinyl Denoising

This section focuses on a real-life example of vinyl denoising. When the vinyl copy of the track Stairway To Heaven was played, an unpleasant tickling sound was corrupting the authentic copy. The desirable outcome of this section is to successfully remove the tick digitally by signal processing.
1. The Vinyl.mat file was downloaded and the matlab command *sound* was used to listen to the track's left and right channels.The tick's nature was regular and periodic. The second channel was noted to be more noisy. It is expected that the tick will appear on the periodogram as a sharp peak in the frequency spectrum. The periodograms of the left and right channel were plotted below.



Correct periodogram plot of the 2 channels

Figure 67: Periodogram of right and left Channel

Looking at the periodograms, it was noted that the two signals look very similar. On a closer look using the *zoom* function it was observed that at $1.5KHz$ for both channels and at around $200Hz$ for the right channel, an irregular peak was present. Further analysis it needed at this point to ensure that the observations are meaningful.

2. Upon listening to the original masterpiece it can be clearly heard that there is a difference from the previous corrupted file. The way to identify the difference between the corrupted and the original signal is by plotting both in the same graph. In this way any mismatching between the two signals will point to the noise tick component.

Correctly explained Tick identification

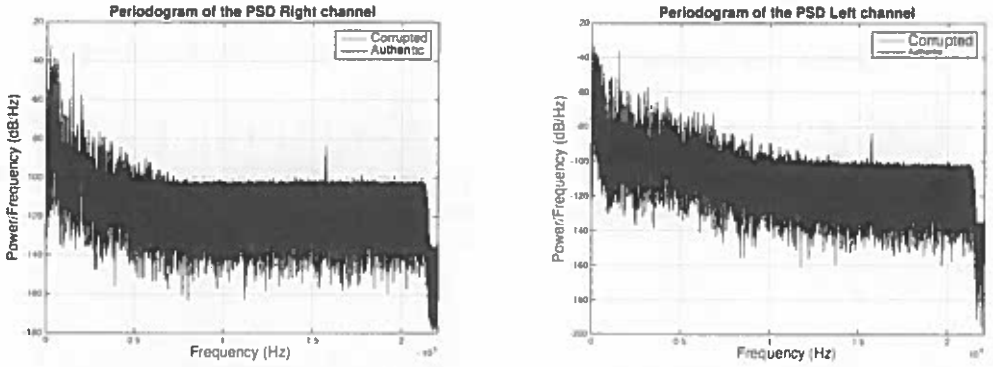


Figure 68: Periodogram of original and corrupted signal plotted together for the right and left Channel

From figure(68) it can be noted that, as predicted before, the two signal are identical except from the red peaks on the left side of the plots. On the left channel the peak was in the range of $1400 - 1600Hz$. The right channel presented two red peaks at $100 - 300Hz$ and $1400 - 1600Hz$ respectively.
3.In this part a fixed-coefficient digital filter was used to remove the distortion. The stop-band butterworth filter was judged to be appropriate due to its capabilities of removing a specific range of frequencies as well as displaying a zero ripple baseband which is particularly useful since it is favourable to remove the noise without corrupting the original signal.The *Matlab* function *butter* was used to apply the filter. The frequencies were normalised given that the *butter* function accepts normalised frequencies. The conversion was done using $f(desirable) = 2f(analogue)/f(sampling)$. After the filter was applied the following plot resulted. The right channel was used to portray the result in figure(69).

Application of notch filter is not stated

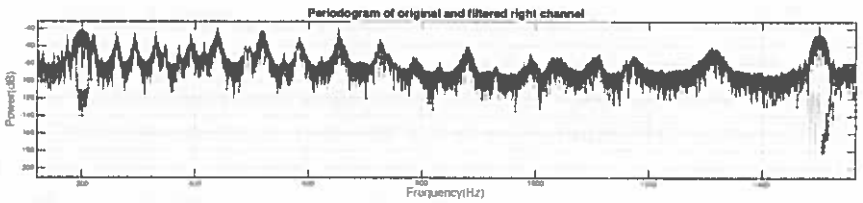


Figure 69: Periodogram of the authentic and corrupted right channel after the butterworth filter was applied

After listening to both channels again it was noted that the filter process was successful and the tick sound had almost completely diminished. A drawback was that the quality of the authentic signal was affected since there are frequencies which are common for both the authentic and the corrupted piece. The filtered signal was

compared to the original signal again using the *sound* command and it was clear that the noise was removed fully. A very small difference with the original piece was present however, which can be justified by the attenuation of some useful components as a result of the noise suppression.
4.Subsequently, the original,corrupted and filtered signals were compared ,using blue red and green signal key respectively, in figure(70) below.

Legend is not included in the plot

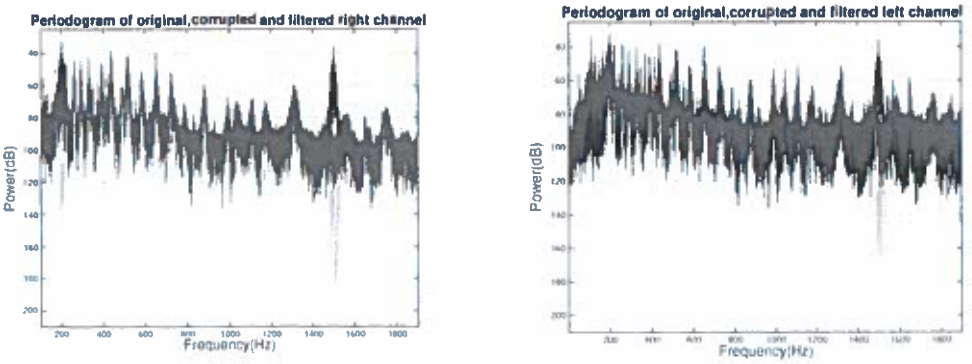


Figure 70: Periodogram of original(red), corrupted(blue) and filtered(green) signal plotted together for the right Channel after butterworth filter

The relative error of the filtered and the corrupted signal was investigated with respect to the ground truth recording, as an extra tool to measure the performance of the signal processing operation. The formula used was:

$$\frac{\|P_c - \hat{P}_c\|}{\|P_c\|} \tag{29}$$

where P_c is is the (non-corrupted) periodogram of channel c and $\hat{P}_c$ is c is your estimate of $Pc, c \in left, right$. Table (9) displays the results

	Left Channel	Right Channel
Filtered	0.0070	0.1075
Corrupted	0.2275	1.3296

Table 9: Relative PSD errors of filtered and corrupted signals

It is evident from table(9) that the filtering action has clearly enhanced the quality of the corrupted signal as seen by the fact that the relative error of the left channel had reduced by nearly 33 times and that of the right channel by 12 times. In order to obtain a more complete conclusion regarding the relative error, its sources should be taken into account. For the case of the filtered signal the error arises from the over-attenuation at the stop-band frequencies, whereas for the case of the corrupted signal error occurs due to the large peaks. The difference in the relative error between the left and the right channel can therefore be explained taking into perspective the 2 unique frequency bands, where the filter error due to the attenuation occurs.

5. The least mean square(LMS) estimator was implemented in this part of the question for the development of a supervised adaptive denoising algorithm,which does not need the analysis of the whole content of the disc. A normalised LMS($nlms$) function was created in *Matlab* and it was preferred from the original *lms* function because if the input of the filter is an audio signal which could potentially become unstable,the output could lead to higher volumes of the audio, giving the listener unnecessary problems. Consequently,a selection of a stable normalised lms would be more appropriate. It is important to note at this point that the *nlms* filter is not indiscriminate in the frequency range that it removes. The *nlms*filter does not over-attenuate at the noise containing frequencies,unlike the stop-band filter. The order of the filter and the learning rate(mu) were varied to investigate the ideal combination which would yield the smallest relative error. Table(10) and (11) below illustrates the different mu and orders tested and the equivalent relative error for the left and right channels.

Sufficient Comparison of the NLMS vs LMS

	p=2	p=4	p=8	p=10	p=50
$\mu = 0.01$	0.0629	0.0583	0.0381	0.0232	0.0011
$\mu = 0.1$	0.0466	0.0399	0.0122	0.0091	0.068
$\mu = 0.5$	0.0361	0.0317	0.0076	0.0062	0.033
$\mu = 1$	0.0443	0.0314	0.0083	0.0094	0.016

Table 10: Relative error of left channel for different μ and orders

	p=2	p=4	p=8	p=10	p=50
$\mu = 0.01$	0.1048	0.1033	0.1021	0.1022	0.0857
$\mu = 0.1$	0.0632	0.0605	0.0672	0.0705	0.0644
$\mu = 0.5$	0.0275	0.0282	0.0364	0.0434	0.0167
$\mu = 1$	0.0211	0.0211	0.0288	0.0363	0.0011

Good use of tables

Table 11: Relative error of right channel for different μ and orders

Looking at the table it can be noted that as the adaptation gain increases, the coefficients appear to rise faster and the error reduces swiftly, repeating the trend obtained in section 4. The ideal values for μ and p for the left channel is 0.5 and 10 respectively and the ideal values for μ and p for the right channel are 0.1 and 2. Eventhough the highest order of 40 produced the best results, it must be taken into account that higher order reflects higher computational complexity. Therefore it would be more economical to use a filter of order 10 and $\mu = 0.3$ for the left channel. Following the same logic, $\mu = 1$ and $p = 2$ were chosen for the right channel. Figure(71) below display the coefficient evolution for the two channel. Computing the *lms* for the same parameters resulted in relative error equivalent to 0.018 and 0.038 for the left and right channel respectively, which is larger compared to the values shown above for the *nlms*.

Legend is small

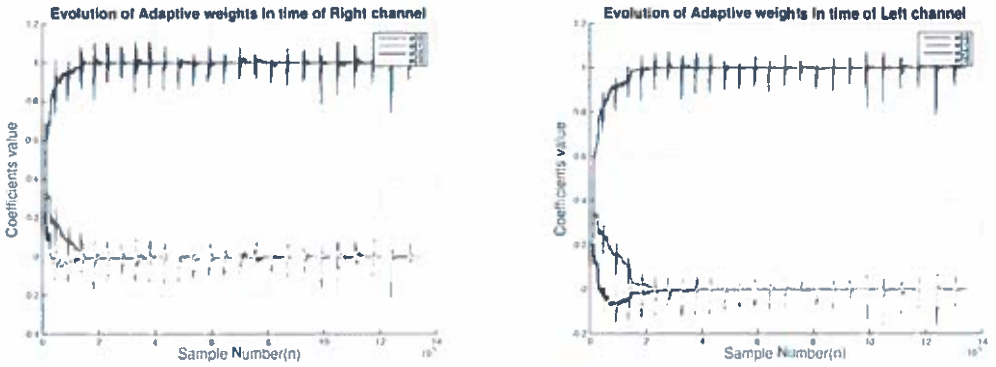


Figure 71: Left and right channel coefficient evolution

6.The LTE version of the masterpiece Stairway-to-heaven with proper rich frequency components was examined in this part of the question. Using the adaptive filter in part5.5 the unwanted signal was removed. Subsequently the *nlms* algorithm was applied to the corrupted version. The same procedure as before was repeated to obtain the optimal learning factor and order. The results were plotted in figure (72) below.

Plot of the Spectral Content of the LTE denoised could've been illustrated separately

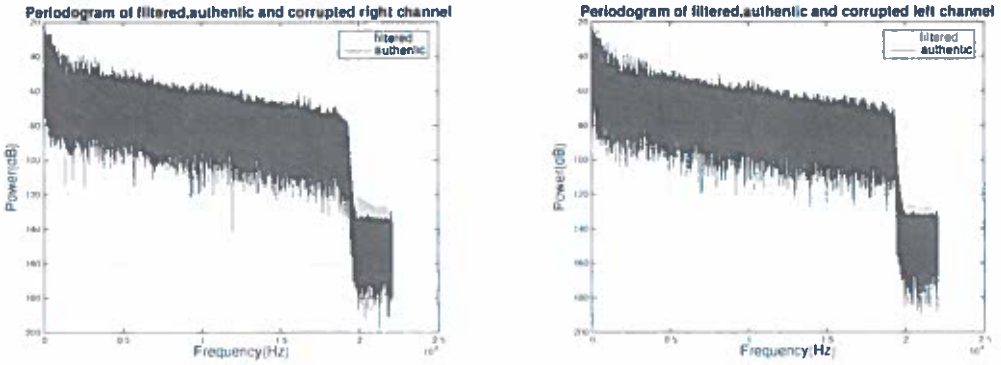


Figure 72: Periodogram of original and corrupted signal plotted together for the right and left Channel for the LTE Stairway to heaven

The optimal combinations of μ and p for the left and the right channel were found to be 0.5, 25 and 1 ,25 respectively. Due to much more sophisticated and higher quality instruments,analogue synthesizers and overlay of different recordings the periodogram appeared much denser with many more frequency components. Furthermore, the LTE version has significant spectral contents in the frequency rand at which the noise occurs and thus attenuating the ticks even slightly has a significant effect on the Signal-to-Noise ratio.This and the nature of the rich frequency signal provided constant hurdles for the adaptive filter in the process of removing the unwanted noise. Having a closer look at the two figures above it was observed that higher frequencies produced a higher error. This is happening because the filter is not allowed enough time to adapt, losing the high frequency components. The relative errors, as expected, appear to be higher than the ones of the lower quality song. The left channel one is 0.0093 and the right channel one is 0.0751. This further strengthens the previous conclusion that the filter experiences some difficulties while

removing the noise. Ultimately, it was concluded that the adaptive filter would be the best choice at this point to produce the best output signal and reduce the tick as much as possible.

References

[1] Hayes,M.H., *Statistical Digital Signal Processing and Modelling*, Willey,1996.

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[3] Mandic, D. D. (2015), *Advanced Signal Processing Lecure Notes.*

[4] Mandic, D. D. (2015), *Advanced Signal Processing Coursework Notes.*